UNIT V TRANSPORT AND APPLICATION LAYER

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5.1 TRANSPORT LAYER

The transport layer is where, as the name implies, data is transported from one host to another. This is where your network really gets moving! The transport layer uses two protocols: TCP and UDP. Think of TCP as getting a registered letter in the mail. You have to sign for it before the mail carrier will let you have it. This slows down the process a bit, but the sender knows for certain that you received the letter and when you received it. UDP is more like a regular, stamped letter. It arrives in your mailbox and, if it does, it is probably intended for you, but it might actually be for someone else who does not live there. Also, it may not arrive in your mailbox at all. The sender cannot be sure you received it. Nevertheless, there are times when UDP, like a stamped letter, is the protocol that is needed. This topic dives into how TCP and UDP work in the transport layer. Later in this module there are several videos to help you understand these processes.

5.1.1 Transportation of Data

Role of the Transport Layer

Application layer programs generate data that must be exchanged between source and destination hosts. The transport layer is responsible for logical communications between applications running on different hosts. This may include services such as establishing a temporary session between two hosts and the reliable transmission of information for an application.

The transport layer has no knowledge of the destination host type, the type of media over which the data must travel, the path taken by the data, the congestion on a link, or the size of the network.

The transport layer includes two protocols:

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)

Transport Layer Responsibilities

The transport layer has many responsibilities.

Select each tab for more information.
Tracking Individual Conversations
Segmenting Data and Reassembling Segments
Add Header Information
Identifying the Applications
Conversation Multiplexing

Tracking Individual Conversations

At the transport layer, each set of data flowing between a source application and a destination application is known as a conversation and is tracked separately. It is the responsibility of the transport layer to maintain and track these multiple conversations.

As illustrated in the figure, a host may have multiple applications that are communicating across the network simultaneously.

Most networks have a limitation on the amount of data that can be included in a single packet. Therefore, data must be divided into manageable pieces.

The PC simultaneously runs multiple network applications including an email client, instant messaging client, web browser web pages, streaming video, and a video conference client.

Email
Online Video Chatting

Streaming Video
Multiple Web Pages
Instant Messaging
Network

Transport Layer Protocols

IP is concerned only with the structure, addressing, and routing of packets. IP does not specify how the delivery or transportation of the packets takes place.

Transport layer protocols specify how to transfer messages between hosts, and are responsible for managing reliability requirements of a conversation. The transport layer includes the TCP and UDP protocols.

Different applications have different transport reliability requirements. Therefore, TCP/IP provides two transport layer protocols, as shown in the figure.

shows how application layer protocols like FTP, HTTP, SMTP use TCP at the transport layer and DNS and TFTP use UDP. How they all use IP at the internet layer regardless of whether they connect to a LAN or a WAN at the network access layer

Application
Transport
Internet
Network Access

FTP
HTTP
(www)
SMTP
(email)
DNS
TFTP
TCP
UDP
IP
LAN
connections
WAN
connections

Transmission Control Protocol (TCP)

IP is concerned only with the structure, addressing, and routing of packets, from original sender to final destination. IP is not responsible for guaranteeing delivery or determining whether a connection between the sender and receiver needs to be established.

TCP is considered a reliable, full-featured transport layer protocol, which ensures that all of the data arrives at the destination. TCP includes fields which ensure the delivery of the application data. These fields require additional processing by the sending and receiving hosts.

Note: TCP divides data into segments.

TCP transport is analogous to sending packages that are tracked from source to destination. If a shipping order is broken up into several packages, a customer can check online to see the order of the delivery.

TCP provides reliability and flow control using these basic operations:

- Number and track data segments transmitted to a specific host from a specific application
- Acknowledge received data
- Retransmit any unacknowledged data after a certain amount of time
- Sequence data that might arrive in wrong order
- Send data at an efficient rate that is acceptable by the receiver

In order to maintain the state of a conversation and track the information, TCP must first establish a connection between the sender and the receiver. This is why TCP is known as a connection-oriented protocol.

The animation shows a connection to an FTP server initiated with a TCP 3-way handshake and the data segments being accounted for by using sequence numbers and acknowledgements

A file is sent to a server using the File Transfer Protocol (FTP) application. TCP tracks the conversation and divides the data to be sent into 6 segments.
The first 3 out of 6 segments are forwarded to the server.
The file server acknowledges the first 3 segments received.
The client forwards the next 3 segments.
No segments are received, no acknowledgement is sent.
The client resends the final 3 segments.
The final 3 segments are received and acknowledged.

ISP 1

FTP

Server Farm

Internet

ISP 2

User Datagram Protocol (UDP)

UDP is a simpler transport layer protocol than TCP. It does not provide reliability and flow control, which means it requires fewer header fields. Because the sender and the receiver UDP processes do not have to manage reliability and flow control, this means UDP datagrams can be processed faster than TCP segments. UDP provides the basic functions for delivering datagrams between the appropriate applications, with very little overhead and data checking.

Note: UDP divides data into datagrams that are also referred to as segments.

UDP is a connectionless protocol. Because UDP does not provide reliability or flow control, it does not require an established connection. Because UDP does not track information sent or received between the client and server, UDP is also known as a stateless protocol.

UDP is also known as a best-effort delivery protocol because there is no acknowledgment that the data is received at the destination. With UDP, there are no transport layer processes that inform the sender of a successful delivery.

UDP is like placing a regular, nonregistered, letter in the mail. The sender of the letter is not aware of the availability of the receiver to receive the letter. Nor is the post office responsible for tracking the letter or informing the sender if the letter does not arrive at the final destination.

A file is sent to a server using the Trivial File Transfer Protocol (TFTP) application. UDP divides the data into datagrams and sends them using best-effort delivery.
The file server receives all 6 segments, no acknowledgment is sent.

ISP 1
TFTP
Server Farm
Internet
ISP 2

The Right Transport Layer Protocol for the Right Application

Some applications can tolerate some data loss during transmission over the network, but delays in transmission are unacceptable. For these applications, UDP is the better choice because it requires less network overhead. UDP is preferable for applications such as Voice over IP (VoIP). Acknowledgments and retransmission would slow down delivery and make the voice conversation unacceptable.

UDP is also used by request-and-reply applications where the data is minimal, and retransmission can be done quickly. For example, Domain Name System (DNS) uses UDP for this type of transaction. The client requests IPv4 and IPv6 addresses for a known domain name from a DNS server. If the client does not receive a response in a predetermined amount of time, it simply sends the request again.

For example, if one or two segments of a live video stream fail to arrive, it creates a momentary disruption in the stream. This may appear as distortion in the image or sound, but may not be noticeable to the user. If the destination device had to account for lost data, the stream could be delayed while waiting for retransmissions, therefore causing the image or sound to be greatly degraded. In this case, it is better to render the best media possible with the segments received, and forego reliability.

For other applications it is important that all the data arrives and that it can be processed in its proper sequence. For these types of applications, TCP is used as the transport protocol. For example, applications such as databases, web browsers, and email clients, require that all data that is sent arrives at the destination in its original condition. Any missing data could corrupt a communication, making it either incomplete or unreadable. For example, it is important when accessing banking information over the web to make sure all the information is sent and received correctly.

Application developers must choose which transport protocol type is appropriate based on the requirements of the applications. Video may be sent over TCP or UDP. Applications that stream stored audio and video typically use TCP. The application uses TCP to perform buffering, bandwidth probing, and congestion control, in order to better control the user experience.

Real-time video and voice usually use UDP, but may also use TCP, or both UDP and TCP. A video conferencing application may use UDP by default, but because many firewalls block UDP, the application can also be sent over TCP.

Applications that stream stored audio and video use TCP. For example, if your network suddenly cannot support the bandwidth needed to watch an on-demand movie, the application pauses the playback. During the pause, you might see a “buffering...” message while TCP works to re-establish the stream. When all the segments are in order and a minimum level of bandwidth is restored, your TCP session resumes, and the movie resumes playing.

The figure summarizes differences between UDP and TCP.

lists the differences between UDP: fast, low overhead, no acknowledgements, no resending and TCP: reliable, acknowledges data, resends lost data, and delivers data with sequence numbers

TCP
UDP

VoIP
(IP telephony)
DNS
(Domain Name Resolution)
SMTP/IMAP
(Email)
HTTP/HTTPS
(World Wide Web

5.1.2 TCP OVERVIEW

TCP Features

In the previous topic, you learned that TCP and UDP are the two transport layer protocols. This topic gives more details about what TCP does and when it is a good idea to use it instead of UDP.

To understand the differences between TCP and UDP, it is important to understand how each protocol implements specific reliability features and how each protocol tracks conversations.

In addition to supporting the basic functions of data segmentation and reassembly, TCP also provides the following services:

- **Establishes a Session** - TCP is a connection-oriented protocol that negotiates and establishes a permanent connection (or session) between source and destination devices prior to forwarding any traffic. Through session establishment, the devices negotiate the amount of traffic that can be forwarded at a given time, and the communication data between the two can be closely managed.
- **Ensures Reliable Delivery** - For many reasons, it is possible for a segment to become corrupted or lost completely, as it is transmitted over the network. TCP ensures that each segment that is sent by the source arrives at the destination.
- **Provides Same-Order Delivery** - Because networks may provide multiple routes that can have different transmission rates, data can arrive in the wrong order. By numbering and sequencing the segments, TCP ensures segments are reassembled into the proper order.
- **Supports Flow Control** - Network hosts have limited resources (i.e., memory and processing power). When TCP is aware that these resources are overtaxed, it can request that the sending application reduce the rate of data flow. This is done by TCP regulating the amount of data the source transmits. Flow control can prevent the need

for retransmission of the data when the resources of the receiving host are overwhelmed.

For more information on TCP, search the internet for the RFC 793.

TCP Header

TCP is a stateful protocol which means it keeps track of the state of the communication session. To track the state of a session, TCP records which information it has sent and which information has been acknowledged. The stateful session begins with the session establishment and ends with the session termination.

TCP Header Fields

The table identifies and describes the ten fields in a TCP header.

TCP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number.
Destination Port	A 16-bit field used to identify the destination application by port number.
Sequence Number	A 32-bit field used for data reassembly purposes.
Acknowledgment Number	A 32-bit field used to indicate that data has been received and the next byte expected from the source.
Header Length	A 4-bit field known as ″data offset″ that indicates the length of the TCP segment header.
Reserved	A 6-bit field that is reserved for future use.
Control bits	A 6-bit field that includes bit codes, or flags, which indicate the purpose and function of the TCP segment.
Window size	A 16-bit field used to indicate the number of bytes that can be accepted at one time.
Checksum	A 16-bit field used for error checking of the segment header and data.
Urgent	A 16-bit field used to indicate if the contained data is urgent.

Applications that use TCP

TCP is a good example of how the different layers of the TCP/IP protocol suite have specific roles. TCP handles all tasks associated with dividing the data stream into segments, providing reliability, controlling data flow, and reordering segments. TCP frees the application from having to manage any of these tasks

5.1.3 UDP Overview

UDP Features

This topic will cover UDP, what it does, and when it is a good idea to use it instead of TCP. UDP is a best-effort transport protocol. UDP is a lightweight transport protocol that offers the same data segmentation and reassembly as TCP, but without TCP reliability and flow control.

UDP is such a simple protocol that it is usually described in terms of what it does not do compared to TCP.

UDP features include the following:

- Data is reconstructed in the order that it is received.
- Any segments that are lost are not resent.
- There is no session establishment.
- The sending is not informed about resource availability.

For more information on UDP, search the internet for the RFC.

UDP Header

UDP is a stateless protocol, meaning neither the client, nor the server, tracks the state of the communication session. If reliability is required when using UDP as the transport protocol, it must be handled by the application.

One of the most important requirements for delivering live video and voice over the network is that the data continues to flow quickly. Live video and voice applications can tolerate some data loss with minimal or no noticeable effect, and are perfectly suited to UDP.

The blocks of communication in UDP are called datagrams, or segments. These datagrams are sent as best effort by the transport layer protocol.

The UDP header is far simpler than the TCP header because it only has four fields and requires 8 bytes (i.e., 64 bits). The figure shows the fields in a UDP header.

UDP datagram diagram shows 4 header fields: source port, destination port, length, and checksum as well as the non header application layer data
8 Bytes

UDP Header Fields

The table identifies and describes the four fields in a UDP header.

UDP Header Field	Description
Source Port	A 16-bit field used to identify the source application by port number.
Destination Port	A 16-bit field used to identify the destination application by port number.
Length	A 16-bit field that indicates the length of the UDP datagram header.
Checksum	A 16-bit field used for error checking of the datagram header and data.

Applications that use UDP

There are three types of applications that are best suited for UDP:

- **Live video and multimedia applications** - These applications can tolerate some data loss, but require little or no delay. Examples include VoIP and live streaming video.
- **Simple request and reply applications** - Applications with simple transactions where a host sends a request and may or may not receive a reply. Examples include DNS and DHCP.
- **Applications that handle reliability themselves** - Unidirectional communications where flow control, error detection, acknowledgments, and error recovery is not required, or can be handled by the application. Examples include SNMP and TFTP.

Although DNS and SNMP use UDP by default, both can also use TCP. DNS will use TCP if the DNS request or DNS response is more than 512 bytes, such as when a DNS response includes many name resolutions. Similarly, under some situations the network administrator may want to configure SNMP to use TCP.

5.1.4 Port Numbers

Multiple Separate Communications

As you have learned, there are some situations in which TCP is the right protocol for the job, and other situations in which UDP should be used. No matter what type of data is being transported, both TCP and UDP use port numbers.

The TCP and UDP transport layer protocols use port numbers to manage multiple, simultaneous conversations. As shown in the figure, the TCP and UDP header fields identify a source and destination application port number.

shows the source port and destination port header fields which are 2 bytes each

The source port number is associated with the originating application on the local host whereas the destination port number is associated with the destination application on the remote host.

For instance, assume a host is initiating a web page request from a web server. When the host initiates the web page request, the source port number is dynamically generated by the host to uniquely identify the conversation. Each request generated by a host will use a different dynamically created source port number. This process allows multiple conversations to occur simultaneously.

In the request, the destination port number is what identifies the type of service being requested of the destination web server.. For example, when a client specifies port 80 in the destination port, the server that receives the message knows that web services are being requested.

A server can offer more than one service simultaneously such as web services on port 80 while it offers File Transfer Protocol (FTP) connection establishment on port 21.

Socket Pairs

The source and destination ports are placed within the segment. The segments are then encapsulated within an IP packet. The IP packet contains the IP address of the source and destination. The combination of the source IP address and source port number, or the destination IP address and destination port number is known as a socket.

In the example in the figure, the PC is simultaneously requesting FTP and web services from the destination server.

The figure depicts a PC making both an FTP connection and a web connection to a server. The requests have source and destination port numbers which identify the host PC and the requested application service respectively.

Dest MAC
Source MAC
Src IP
Dest. IP
Source Port
Dest. Port

FTP client Source Port: 1305
Web Client Source Port: 1099
FTP Server Dest Port: 21
Web Server Dest Port: 80
FTP Connection
Web Connection
Dest MAC
Source MAC
Src. IP
Dest IP
Source
Port
Dest.
Port
Source
192.168.1.5
00-07-E9-63-CE-53
Destination
192.168.1.7
00-07-E9-42-AC-28
FTP
Web

In the example, the FTP request generated by the PC includes the Layer 2 MAC addresses and the Layer 3 IP addresses. The request also identifies the source port number 1305 (i.e., dynamically generated by the host) and destination port, identifying the FTP services on port 21. The host also has requested a web page from the server using the same Layer 2 and Layer 3 addresses. However, it is using the source port number 1099 (i.e., dynamically generated by the host) and destination port identifying the web service on port 80.

The socket is used to identify the server and service being requested by the client. A client socket might look like this, with 1099 representing the source port number: 192.168.1.5:1099

The socket on a web server might be 192.168.1.7:80

Together, these two sockets combine to form a *socket pair*: 192.168.1.5:1099, 192.168.1.7:80

Sockets enable multiple processes, running on a client, to distinguish themselves from each other, and multiple connections to a server process to be distinguished from each other.

The source port number acts as a return address for the requesting application. The transport layer keeps track of this port and the application that initiated the request so that when a response is returned, it can be forwarded to the correct application.

Port Number Groups

The Internet Assigned Numbers Authority (IANA) is the standards organization responsible for assigning various addressing standards, including the 16-bit port numbers. The 16 bits used to identify the source and destination port numbers provides a range of ports from 0 through 65535.

The IANA has divided the range of numbers into the following three port groups.

Port Group	Number Range	Description

Well-known Ports	**0 to 1,023**	• These port numbers are reserved for common or popular services and applications such as web browsers, email clients, and remote access clients. • Defined well-known ports for common server applications enables clients to easily identify the associated service required.
Registered Ports	**1,024 to 49,151**	• These port numbers are assigned by IANA to a requesting entity to use with specific processes or applications. • These processes are primarily individual applications that a user has chosen to install, rather than common applications that would receive a well-known port number. • For example, Cisco has registered port 1812 for its RADIUS server authentication process.
Private and/or **Dynamic Ports**	**49,152 to 65,535**	• These ports are also known as *ephemeral ports*. • The client's OS usually assign port numbers dynamically when a connection to a service is initiated. • The dynamic port is then used to identify the client application during communication.

Note: Some client operating systems may use registered port numbers instead of dynamic port numbers for assigning source ports.

The table displays some common well-known port numbers and their associated applications.

Well-Known Port Numbers

Port Number	Protocol	Application
20	TCP	File Transfer Protocol (FTP) - Data
21	TCP	File Transfer Protocol (FTP) - Control
22	TCP	Secure Shell (SSH)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	UDP, TCP	Domain Name System (DNS)
67	UDP	Dynamic Host Configuration Protocol (DHCP) - Server
68	UDP	Dynamic Host Configuration Protocol - Client
69	UDP	Trivial File Transfer Protocol (TFTP)

80	TCP	Hypertext Transfer Protocol (HTTP)
110	TCP	Post Office Protocol version 3 (POP3)
143	TCP	Internet Message Access Protocol (IMAP)
161	UDP	Simple Network Management Protocol (SNMP)
443	TCP	Hypertext Transfer Protocol Secure (HTTPS)

Some applications may use both TCP and UDP. For example, DNS uses UDP when clients send requests to a DNS server. However, communication between two DNS servers always uses TCP.

Search the IANA website for port registry to view the full list of port numbers and associated applications.

14.4.4 The netstat Command

Unexplained TCP connections can pose a major security threat. They can indicate that something or someone is connected to the local host. Sometimes it is necessary to know which active TCP connections are open and running on a networked host. Netstat is an important network utility that can be used to verify those connections. As shown below, enter the command **netstat** to list the protocols in use, the local address and port numbers, the foreign address and port numbers, and the connection state.

```
C:\> netstat
Active Connections
  Proto  Local Address          Foreign
Address            State
  TCP    192.168.1.124:3126     192.168.0.2:netbios-
ssn    ESTABLISHED
  TCP    192.168.1.124:3158     207.138.126.152:http       ESTA
BLISHED
  TCP    192.168.1.124:3159     207.138.126.169:http       ESTA
BLISHED
  TCP    192.168.1.124:3160     207.138.126.169:http       ESTA
BLISHED
  TCP    192.168.1.124:3161     sc.msn.com:http            ESTA
BLISHED
  TCP    192.168.1.124:3166     www.cisco.com:http         ESTA
BLISHED
(output omitted)
C:\>
```

By default, the **netstat** command will attempt to resolve IP addresses to domain names and port numbers to well-known applications. The **-n** option can be used to display IP addresses and port numbers in their numerical form.

5.1.5 TCP COMMUNICATION PROCESS

TCP Server Processes

You already know the fundamentals of TCP. Understanding the role of port numbers will help you to grasp the details of the TCP communication process. In this topic, you will also learn about the TCP three-way handshake and session termination processes.

Each application process running on a server is configured to use a port number. The port number is either automatically assigned or configured manually by a system administrator.

An individual server cannot have two services assigned to the same port number within the same transport layer services. For example, a host running a web server application and a file transfer application cannot have both configured to use the same port, such as TCP port 80.

An active server application assigned to a specific port is considered open, which means that the transport layer accepts, and processes segments addressed to that port. Any incoming client request addressed to the correct socket is accepted, and the data is passed to the server application. There can be many ports open simultaneously on a server, one for each active server application.

Select each tab for more information about TCP server processes.
Clients Sending TCP Requests
Request Destination Ports
Request Source Ports
Response Destination Ports
Response Source Ports

Clients Sending TCP Requests

Client 1 is requesting web services and Client 2 is requesting email service of the same sever.

HTTP Request:
Source Port: 49152
Destination Port: 80
SMTP Request:
Source Port: 51152
Destination Port: 25
Server
Client 1
Client 2
HTTP: Port 80
SMTP: Port 25
Client requests to TCP server

TCP Connection Establishment

In some cultures, when two persons meet, they often greet each other by shaking hands. Both parties understand the act of shaking hands as a signal for a friendly greeting. Connections on the network are similar. In TCP connections, the host client establishes the connection with the server using the three-way handshake process.

Select each tab for more information about each TCP connection establishment step.
Step 1. SYN
Step 2. ACK and SYN
Step 3. ACK

Step 1. SYN

The initiating client requests a client-to-server communication session with the server.

PCA initiates a three way handshake by sending a syn segment to PCB.
1AB

Send SYN

(SEQ=100 CTL=SYN)

SYN received

A sends SYN request to B

The three-way handshake validates that the destination host is available to communicate. In this example, host A has validated that host B is available.

Session Termination

To close a connection, the Finish (FIN) control flag must be set in the segment header. To end each one-way TCP session, a two-way handshake, consisting of a FIN segment and an Acknowledgment (ACK) segment, is used. Therefore, to terminate a single conversation supported by TCP, four exchanges are needed to end both sessions. Either the client or the server can initiate the termination.

In the example, the terms client and server are used as a reference for simplicity, but any two hosts that have an open session can initiate the termination process.

Select each tab for more information about the session termination steps.
Step 1. FIN
Step 2. ACK
Step 3. FIN
Step 4. ACK

Step 1. FIN

When the client has no more data to send in the stream, it sends a segment with the FIN flag set.

PCA sends a fin segment to PCB to end the session when there is no more data to send
1AB

Send FIN

FIN received

A sends FIN request to B.

When all segments have been acknowledged, the session is closed.

TCP Three-way Handshake Analysis

Hosts maintain state, track each data segment within a session, and exchange information about what data is received using the information in the TCP header. TCP is a full-duplex protocol, where each connection represents two one-way communication sessions. To establish the connection, the hosts perform a three-way handshake. As shown in the figure, control bits in the TCP header indicate the progress and status of the connection.

These are the functions of the three-way handshake:

- It establishes that the destination device is present on the network.
- It verifies that the destination device has an active service and is accepting requests on the destination port number that the initiating client intends to use.
- It informs the destination device that the source client intends to establish a communication session on that port number.

After the communication is completed the sessions are closed, and the connection is terminated. The connection and session mechanisms enable TCP reliability function.

Control Bits Field

shows the tcp segment header fields with the control bits field of 6 bits highlighted

20 Bytes

The six bits in the Control Bits field of the TCP segment header are also known as flags. A flag is a bit that is set to either on or off.

The six control bits flags are as follows:

- **URG** - Urgent pointer field significant
- **ACK** - Acknowledgment flag used in connection establishment and session termination
- **PSH** - Push function
- **RST** - Reset the connection when an error or timeout occurs
- **SYN** - Synchronize sequence numbers used in connection establishment
- **FIN** - No more data from sender and used in session termination

Search the internet to learn more about the PSH and URG flags.

5.1.6 Reliability and Flow Control

TCP Reliability - Guaranteed and Ordered Delivery

The reason that TCP is the better protocol for some applications is because, unlike UDP, it resends dropped packets and numbers packets to indicate their proper order before delivery. TCP can also help maintain the flow of packets so that devices do not become overloaded. This topic covers these features of TCP in detail.

There may be times when TCP segments do not arrive at their destination. Other times, the TCP segments might arrive out of order. For the original message to be understood by the recipient, all the data must be received and the data in these segments must be reassembled into the original order. Sequence numbers are assigned in the header of each packet to achieve this goal. The sequence number represents the first data byte of the TCP segment.

During session setup, an initial sequence number (ISN) is set. This ISN represents the starting value of the bytes that are transmitted to the receiving application. As data is transmitted during the session, the sequence number is incremented by the number of bytes that have been transmitted. This data byte tracking enables each segment to be uniquely identified and acknowledged. Missing segments can then be identified.

The ISN does not begin at one but is effectively a random number. This is to prevent certain types of malicious attacks. For simplicity, we will use an ISN of 1 for the examples in this chapter.

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TCP Reliability - Data Loss and Retransmission

No matter how well designed a network is, data loss occasionally occurs. TCP provides methods of managing these segment losses. Among these is a mechanism to retransmit segments for unacknowledged data.

The sequence (SEQ) number and acknowledgement (ACK) number are used together to confirm receipt of the bytes of data contained in the transmitted segments. The SEQ number identifies the first byte of data in the segment being transmitted. TCP uses the ACK number sent back to the source to indicate the next byte that the receiver expects to receive. This is called expectational acknowledgement.

Prior to later enhancements, TCP could only acknowledge the next byte expected. For example, in the figure, using segment numbers for simplicity, host A sends segments 1 through 10 to host B. If all the segments arrive except for segments 3 and 4, host B would reply with acknowledgment specifying that the next segment expected is segment 3. Host A has no idea if any other segments arrived or not. Host A would, therefore, resend segments 3 through 10. If all the resent segments arrived successfully, segments 5 through 10 would be duplicates. This can lead to delays, congestion, and inefficiencies.

shows PCA sending 10 segments to PCB, but segments 3 and 4 fail to arrive. So starting with segment 3, PCA resends segments 3 through 10, even though PCB only needed segments 3 and 4
ABAB
Segment 1
Segment 2
Segment 3
Segment 5
Segment 6
Segment 7
Segment 8
Segment 9
Segment 10
Segment 4
Segment 3
Segment 4
Segment 5
Segment 6
Segment 7
Segment 8
Segment 9
Segment 10
ACK 3
ACK 11
Duplicate
segments

Host operating systems today typically employ an optional TCP feature called selective acknowledgment (SACK), negotiated during the three-way handshake. If both hosts support SACK, the receiver can explicitly acknowledge which segments (bytes) were received including any discontinuous segments. The sending host would therefore only need to retransmit the missing data. For example, in the next figure, again using segment numbers for simplicity, host A sends segments 1 through 10 to host B. If all the segments arrive except for segments 3 and 4, host B can acknowledge that it has received segments 1 and 2 (ACK 3), and selectively acknowledge segments 5 through 10 (SACK 5-10). Host A would only need to resend segments 3 and 4.

show PCA sending 10 segments to PCB, but segments 3 and 4 fail to arrive. This time PCB sends an ack 3 and a sack 5-10 letting PCA know to resend missing segments 3 and 4 and the continue with segment 11
ABAB
Segment 1
Segment 2
Segment 3
Segment 5
Segment 6
Segment 7
Segment 8
Segment 9
Segment 10
Segment 4
Segment 3
Segment 4
Segment 11
Segment 12
ACK 3,SACK 5-10
ACK 13

Note: TCP typically sends ACKs for every other packet, but other factors beyond the scope of this topic may alter this behavior.

TCP uses timers to know how long to wait before resending a segment. In the figure, play the video and click the link to download the PDF file. The video and PDF file examine TCP data loss and retransmission.

TCP Flow Control - Window Size and Acknowledgments

TCP also provides mechanisms for flow control. Flow control is the amount of data that the destination can receive and process reliably. Flow control helps maintain the reliability of TCP transmission by adjusting the rate of data flow between source and destination for a given session. To accomplish this, the TCP header includes a 16-bit field called the window size.

.

TCP Window Size Example

shows PCB sending PCA a negotiated window size of 10,000 bytes and a maximum segment size of 1,460 bytes. PCA starts sending segments starting with sequence number 1. An acknowledgement from PCB can be sent without waiting until the window size is reached and the window size can be adjusted by PCA creating a sliding window
AB

MSS = Maximum Segment Size

During three-way handshake Window size 10,000, MSS 1,460

Sequence number 1,461

1,460 bytes
ACK 2,921
Window size 10,000
1,460 bytes
ACK 4,381
Window size 10,000

Send window 10,000
Sequence number 1

1,460 bytes

Receive acknowledgement
Send window 12,920
Sequence number 2,921
Receive acknowledgement
Send window 14,380

Receive 1 – 1,460
Receive 1,461 – 2,920
Receive 2,921 – 4,380

The window size determines the number of bytes that can be sent before expecting an acknowledgment. The acknowledgment number is the number of the next expected byte.

The window size is the number of bytes that the destination device of a TCP session can accept and process at one time. In this example, the PC B initial window size for the TCP session is 10,000 bytes. Starting with the first byte, byte number 1, the last byte PC A can send without receiving an acknowledgment is byte 10,000. This is known as the send window of PC A. The window size is included in every TCP segment so the destination can modify the window size at any time depending on buffer availability.

The initial window size is agreed upon when the TCP session is established during the three-way handshake. The source device must limit the number of bytes sent to the destination device based on the window size of the destination. Only after the source device receives an acknowledgment that the bytes have been received, can it continue sending more data for the session. Typically, the destination will not wait for all the bytes for its window size to be received before replying with an acknowledgment. As the bytes are received and processed, the destination will send acknowledgments to inform the source that it can continue to send additional bytes.

For example, it is typical that PC B would not wait until all 10,000 bytes have been received before sending an acknowledgment. This means PC A can adjust its send window as it receives acknowledgments from PC B. As shown in the figure, when PC A receives an acknowledgment with the acknowledgment number 2,921, which is the next expected byte. The PC A send window will increment 2,920 bytes. This changes the send window from 10,000 bytes to 12,920. PC A can now continue to send up to another 10,000 bytes to PC B as long as it does not send more than its new send window at 12,920.

A destination sending acknowledgments as it processes bytes received, and the continual adjustment of the source send window, is known as sliding windows. In the previous example, the send window of PC A increments or slides over another 2,921 bytes from 10,000 to 12,920.

If the availability of the destination’s buffer space decreases, it may reduce its window size to inform the source to reduce the number of bytes it should send without receiving an acknowledgment.

Note: Devices today use the sliding windows protocol. The receiver typically sends an acknowledgment after every two segments it receives. The number of segments received before being acknowledged may vary. The advantage of sliding windows is that it allows the sender to continuously transmit segments, as long as the receiver is acknowledging previous segments. The details of sliding windows are beyond the scope of this course.

TCP Flow Control - Maximum Segment Size (MSS)

In the figure, the source is transmitting 1,460 bytes of data within each TCP segment. This is typically the Maximum Segment Size (MSS) that the destination device can receive. The MSS is part of the options field in the TCP header that specifies the largest amount of data, in bytes, that a device can receive in a single TCP segment. The MSS size does not include the TCP header. The MSS is typically included during the three-way handshake.

shows the same diagram as before but the emphasis is on the MSS of maximum segment size of 1460 AB

MSS = Maximum Segment Size

During three-way handshake Window size 10,000, MSS 1,460

Sequence number 1,461

1,460 bytes
ACK 2,921
Window size 10,000
1,460 bytes
ACK 4,381
Window size 10,000

Send window 10,000
Sequence number 1

1,460 bytes

Receive acknowledgement
Send window 12,920
Sequence number 2,921
Receive acknowledgement
Send window 14,380

Receive 1 – 1,460
Receive 1,461 – 2,920
Receive 2,921 – 4,380

A common MSS is 1,460 bytes when using IPv4. A host determines the value of its MSS field by subtracting the IP and TCP headers from the Ethernet maximum transmission unit (MTU). On an Ethernet interface, the default MTU is 1500 bytes. Subtracting the IPv4 header of 20 bytes and the TCP header of 20 bytes, the default MSS size will be 1460 bytes, as shown in the figure.

shows a diagram of an entire Ethernet frame of which the MTU is 1500 bytes, with 20 bytes being the IP header, and 20 bytes being the TCP header, this leaves 1460 bytes which is the TCP maximum segment size MSS

Ethernet MTU
IP MTU
20 bytes
20 bytes
1460 bytes
1500 bytes
TCP MSS

TCP Flow Control - Congestion Avoidance

When congestion occurs on a network, it results in packets being discarded by the overloaded router. When packets containing TCP segments do not reach their destination, they are left unacknowledged. By determining the rate at which TCP segments are sent but not acknowledged, the source can assume a certain level of network congestion.

Whenever there is congestion, retransmission of lost TCP segments from the source will occur. If the retransmission is not properly controlled, the additional retransmission of the TCP segments can make the congestion even worse. Not only are new packets with TCP segments introduced into the network, but the feedback effect of the retransmitted TCP segments that were lost will also add to the congestion. To avoid and control congestion, TCP employs several congestion handling mechanisms, timers, and algorithms.

If the source determines that the TCP segments are either not being acknowledged or not acknowledged in a timely manner, then it can reduce the number of bytes it sends before receiving an acknowledgment. As illustrated in the figure, PC A senses there is congestion and therefore, reduces the number of bytes it sends before receiving an acknowledgment from PC B.

TCP Congestion Control

shows PCA sending segments to PCB where lost segments and retransmission can cause congestion

AB

TCP segment 1

TCP segment 2

TCP segment 3

TCP segment 4

Acknowledgement segment 2

TCP segment 2

TCP segment 3

I'm not getting the acknowledgments I expect from PC B so I will reduce the number of bytes I send before getting an acknowledgement.

Acknowledgment numbers are for the next expected byte and not for a segment. The segment numbers used are simplified for illustration purposes.

Notice that it is the source that is reducing the number of unacknowledged bytes it sends and not the window size determined by the destination.

Note: Explanations of actual congestion handling mechanisms, timers, and algorithms are beyond the scope of this course.

5.1.7 UDP COMMUNICATION

UDP Low Overhead versus Reliability

As explained before, UDP is perfect for communications that need to be fast, like VoIP. This topic explains in detail why UDP is perfect for some types of transmissions. As shown in the figure, UDP does not establish a connection. UDP provides low overhead data transport because it has a small datagram header and no network management traffic.

shows a host sender needing to send voice and video data which is sent with UDP which requires no prior negotiated connection

Network
Sender
Receiver
Data
UDP does not establish a connection before sending data.

UDP Datagram Reassembly

Like segments with TCP, when UDP datagrams are sent to a destination, they often take different paths and arrive in the wrong order. UDP does not track sequence numbers the way TCP does. UDP has no way to reorder the datagrams into their transmission order, as shown in the figure.

Therefore, UDP simply reassembles the data in the order that it was received and forwards it to the application. If the data sequence is important to the application, the application must identify the proper sequence and determine how the data should be processed.

UDP: Connectionless and Unreliable

shows UDP datagrams being sent in order but arriving out of order due to the possibility of different routes to reach the destination

Having taken different routes to the destination, datagrams arrive out of order.
Out of order datagrams are not re-ordered.
Different datagrams may take different routes.
Lost datagrams are not re-sent.

UDP Server Processes and Requests

Like TCP-based applications, UDP-based server applications are assigned well-known or registered port numbers, as shown in the figure. When these applications or processes are running on a server, they accept the data matched with the assigned port number. When UDP receives a datagram destined for one of these ports, it forwards the application data to the appropriate application based on its port number.

UDP Server Listening for Requests

shows that a RADIUS server application uses UDP to listen for requests on port 53

Server
Client 1
Client 2
Server Applications
Client DNS requests will be received on Port 53.

Client RADIUS requests will be received on Port 1812.
DNS request
RADIUS request

Note: The Remote Authentication Dial-in User Service (RADIUS) server shown in the figure provides authentication, authorization, and accounting services to manage user access. The operation of RADIUS is beyond the scope for this course.

UDP Client Processes

As with TCP, client-server communication is initiated by a client application that requests data from a server process. The UDP client process dynamically selects a port number from the range of port numbers and uses this as the source port for the conversation. The destination port is usually the well-known or registered port number assigned to the server process.

After a client has selected the source and destination ports, the same pair of ports are used in the header of all datagrams in the transaction. For the data returning to the client from the server, the source and destination port numbers in the datagram header are reversed.

Select each tab for an illustration of two hosts requesting services from the DNS and RADIUS authentication server.
Clients Sending UDP Requests
UDP Request Destination Ports
UDP Request Source Ports
UDP Response Destination
UDP Response Source Ports

Clients Sending UDP Requests

Client 1 is sending a DNS request while Client 2 is requesting RADIUS authentication services of the same server.

Two different PC clients need to make a request to a DNS server
Client 1 DNS Request:
Source Port 49152
Destination Port 53
Client 2 RADIUS User Authentication Request:
Source Port 51152
Destination Port 1812

5.2 APPLICATION LAYER

5.2.1 Application, Presentation and Session Layer

Application Layer

In the OSI and the TCP/IP models, the application layer is the closest layer to the end user. As shown in the figure, it is the layer that provides the interface between the applications used to communicate, and the underlying network over which messages are transmitted. Application

layer protocols are used to exchange data between programs running on the source and destination hosts.

Based on the TCP/IP model, the upper three layers of the OSI model (application, presentation, and session) define functions of the TCP/IP application layer.

There are many application layer protocols, and new protocols are always being developed. Some of the most widely known application layer protocols include Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP), Internet Message Access Protocol (IMAP), and Domain Name System (DNS) protocol.

Presentation and Session Layer

Presentation Layer

The presentation layer has three primary functions:

- Formatting, or presenting, data at the source device into a compatible format for receipt by the destination device.
- Compressing data in a way that can be decompressed by the destination device.
- Encrypting data for transmission and decrypting data upon receipt.

Session Layer

As the name implies, functions at the session layer create and maintain dialogs between source and destination applications. The session layer handles the exchange of information to initiate dialogs, keep them active, and to restart sessions that are disrupted or idle for a long period of time.

TCP/IP Application Layer Protocols

The TCP/IP application protocols specify the format and control information necessary for many common internet communication functions. Application layer protocols are used by both the source and destination devices during a communication session. For the communications to be successful, the application layer protocols that are implemented on the source and destination host must be compatible.

Click each application protocol type to learn more about each protocol.
Name System
Host Config
Email
File Transfer
Web

Name System

DNS - Domain Name System (or Service)

- TCP, UDP 53
- Translates domain names, such as cisco.com, into IP addresses.

5.2.2 Peer-to-Peer

Client-Server Model

In the previous topic, you learned that TCP/IP application layer protocols implemented on both the source and destination host must be compatible. In this topic you will learn about the client/server model and the processes used, which are in the application layer. The same is true for a peer-to-peer network. In the client/server model, the device requesting the information is called a client and the device responding to the request is called a server. The client is a hardware/software combination that people use to directly access the resources that are stored on the server.

Client and server processes are considered to be in the application layer. The client begins the exchange by requesting data from the server, which responds by sending one or more streams of data to the client. Application layer protocols describe the format of the requests and responses between clients and servers. In addition to the actual data transfer, this exchange may also require user authentication and the identification of a data file to be transferred.

One example of a client/server network is using the email service of an ISP to send, receive, and store email. The email client on a home computer issues a request to the email server of the ISP for any unread mail. The server responds by sending the requested email to the client. Data transfer from a client to a server is referred to as an upload and data from a server to a client as a download.

As shown in the figure, files are downloaded from the server to the client.

The figure depicts the client server model. A client which is depicted as a cell phone, computer, or VoIP phone is connected to a server and downloading files from the server.

Download
Network
Client
Server

Peer-to-Peer Networks

In the peer-to-peer (P2P) networking model, the data is accessed from a peer device without the use of a dedicated server.

The P2P network model involves two parts: P2P networks and P2P applications. Both parts have similar features, but in practice work quite differently.

In a P2P network, two or more computers are connected via a network and can share resources (such as printers and files) without having a dedicated server. Every connected end

device (known as a peer) can function as both a server and a client. One computer might assume the role of server for one transaction while simultaneously serving as a client for another. The roles of client and server are set on a per request basis.

In addition to sharing files, a network such as this one would allow users to enable networked games or share an internet connection.

In a peer-to-peer exchange, both devices are considered equal in the communication process. Peer 1 has files that are shared with Peer 2 and can access the shared printer that is directly connected to Peer 2 to print files. Peer 2 is sharing the directly connected printer with Peer 1 while accessing the shared files on Peer 1, as shown in the figure.

The figure depicts the peer to peer model. Two computers are connected by a switch and are communicating directly with each other. In addition a printer is shared by one of the computers and can be accessed by either computer in the figure.

Peer 1
Print Client
File Server
Peer 2
Directly connected printer
Printer
Print Server
File Client

Peer-to-Peer Applications

A P2P application allows a device to act as both a client and a server within the same communication, as shown in the figure. In this model, every client is a server and every server is a client. P2P applications require that each end device provide a user interface and run a background service.

Some P2P applications use a hybrid system where resource sharing is decentralized, but the indexes that point to resource locations are stored in a centralized directory. In a hybrid system, each peer accesses an index server to get the location of a resource stored on another peer.

The figure depicts a Instant Message conversation with two machines communicating with each other through the network. Each machine is acting as both a client and server. The caption states Both clients can simultaneously send and receive messages.

Good.
I'll be there.
Meeting tonight.
I'll be there.
Good.
Meeting tonight.
Client and Server
Network
Client and Server

Both clients can simultaneously send and receive messages.

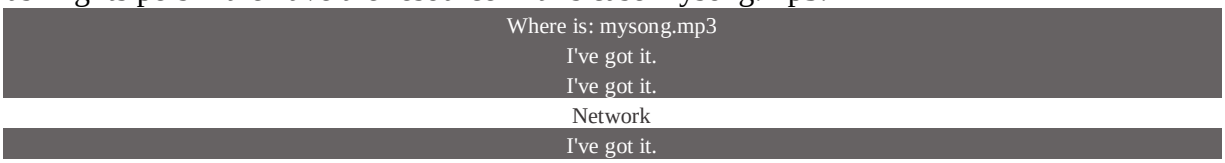
Common P2P Applications

With P2P applications, each computer in the network that is running the application can act as a client or a server for the other computers in the network that are also running the application. Common P2P networks include the following:

- BitTorrent
- Direct Connect
- eDonkey
- Freenet

Some P2P applications are based on the Gnutella protocol, where each user shares whole files with other users. As shown in the figure, Gnutella-compatible client software allows users to connect to Gnutella services over the internet, and to locate and access resources shared by other Gnutella peers. Many Gnutella client applications are available, including µTorrent, BitComet, DC++, Deluge, and emule.

The figure shows a P2P application searching for shared resources. The P2papplication is asking its pers if the have the resource in this case mysong.mp3.



Gnutella P2P applications search for shared resources on multiple peers.

Many P2P applications allow users to share pieces of many files with each other at the same time. Clients use a torrent file to locate other users who have pieces that they need so that they can then connect directly to them. This file also contains information about tracker computers that keep track of which users have specific pieces of certain files. Clients ask for pieces from multiple users at the same time. This is known as a swarm and the technology is called BitTorrent. BitTorrent has its own client. But there are many other BitTorrent clients including uTorrent, Deluge, and qBittorrent.

Note: Any type of file can be shared between users. Many of these files are copyrighted, meaning that only the creator has the right to use and distribute them. It is against the law to download or distribute copyrighted files without permission from the copyright holder. Copyright violation can result in criminal charges and civil lawsuits.

5.2.3 Web and Email Protocols

Hypertext Transfer Protocol and Hypertext Markup Language

There are application layer-specific protocols that are designed for common uses such as web browsing and email. The first topic gave you an overview of these protocols. This topic goes into more detail.

When a web address or Uniform Resource Locator (URL) is typed into a web browser, the web browser establishes a connection to the web service. The web service is running on the server that is using the HTTP protocol. URLs and Uniform Resource Identifiers (URIs) are the names most people associate with web addresses.

To better understand how the web browser and web server interact, examine how a web page is opened in a browser. For this example, use the http://www.cisco.com/index.html URL.

Click each button for more information.
Step 1
Step 2
Step 3
Step 4

Step 1

The browser interprets the three parts of the URL:

- http (the protocol or scheme)
- www.cisco.com (the server name)
- index.html (the specific filename requested)

shows a http server connected thru the internet to a client computer. The client computer is requesting the url www.cisco.com which is the HTTP server.

HTTP Server
Network
Client
http://www.cisco.com/index.html

HTTP and HTTPS

HTTP is a request/response protocol. When a client, typically a web browser, sends a request to a web server, HTTP specifies the message types used for that communication. The three common message types are GET (see figure), POST, and PUT:

- **GET** - This is a client request for data. A client (web browser) sends the GET message to the web server to request HTML pages.
- **POST** - This uploads data files to the web server, such as form data.
- **PUT** - This uploads resources or content to the web server, such as an image.

The figure depicts a client performing a HTTP request to a HTTP server. THe file requested is a Fully Qualified Domain Name. The request uses a Get to retrieve the web page. The URL field is shown on the client computer as a http://www.cisco.com request.

HTTP Server
Client
Uniform Resource Locator (URL)
HTTP Request

File Requested
Host : www.cisco.com
GET /index.html HTTP/1.1
Fully qualified domain name

Although HTTP is remarkably flexible, it is not a secure protocol. The request messages send information to the server in plaintext that can be intercepted and read. The server responses, typically HTML pages, are also unencrypted.

For secure communication across the internet, the HTTP Secure (HTTPS) protocol is used. HTTPS uses authentication and encryption to secure data as it travels between the client and server. HTTPS uses the same client request-server response process as HTTP, but the data stream is encrypted with Transport Layer Security (TLS) or its predecessor Secure Socket Layer (SSL) before being transported across the network.

Email Protocols

One of the primary services offered by an ISP is email hosting. To run on a computer or other end device, email requires several applications and services, as shown in the figure. Email is a store-and-forward method of sending, storing, and retrieving electronic messages across a network. Email messages are stored in databases on mail servers.

The figure depicts an email transaction from a sender using the SMTP protocol sending an email to recipient@cisco.com through an I S P mail server A arriving at the recipients I S P mail server B and the recipient reading the email using either an I MAP or POP protocol.

Internet

IMAP or POP3

SMTP

ISP A

ISP B

Sender

Recipient

Mail Server

Mail Server

SMTP

Send to
recipient@cisco.com

Email clients communicate with mail servers to send and receive email. Mail servers communicate with other mail servers to transport messages from one domain to another. An email client does not communicate directly with another email client when sending email. Instead, both clients rely on the mail server to transport messages.

Email supports three separate protocols for operation: Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP), and IMAP. The application layer process that sends mail uses SMTP. A client retrieves email using one of the two application layer protocols: POP or IMAP.

15.3.4 SMTP, POP, and IMAP
Click each button for more information.
SMTP
POP
IMAP

SMTP

SMTP message formats require a message header and a message body. Although the message body can contain any amount of text, the message header must have a properly formatted recipient email address and a sender address.

When a client sends email, the client SMTP process connects with a server SMTP process on well-known port 25. After the connection is made, the client attempts to send the email to the server across the connection. When the server receives the message, it either places the message in a local account, if the recipient is local, or forwards the message to another mail server for delivery.

The destination email server may not be online, or may be busy, when email messages are sent. Therefore, SMTP spools messages to be sent at a later time. Periodically, the server checks the queue for messages and attempts to send them again. If the message is still not delivered after a predetermined expiration time, it is returned to the sender as undeliverable.

5.2.4 IP Addressing Services

Domain Name System

There are other application layer-specific protocols that were designed to make it easier to obtain addresses for network devices. These services are essential because it would be very time consuming to remember IP addresses instead of URLs or manually configure all of the devices in a medium to large network. The first topic in this module gave you an overview of these protocols. This topic goes into more detail about the IP addressing services, DNS and DHCP.

In data networks, devices are labeled with numeric IP addresses to send and receive data over networks. Domain names were created to convert the numeric address into a simple, recognizable name.

On the internet, fully-qualified domain names (FQDNs), such as http://www.cisco.com, are much easier for people to remember than 198.133.219.25, which is the actual numeric address for this server. If Cisco decides to change the numeric address of www.cisco.com, it is transparent to the user because the domain name remains the same. The new address is simply linked to the existing domain name and connectivity is maintained.

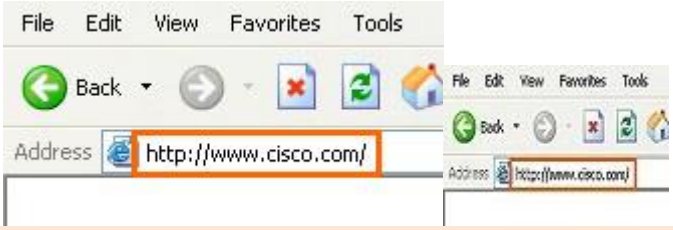
The DNS protocol defines an automated service that matches resource names with the required numeric network address. It includes the format for queries, responses, and data. The DNS protocol communications use a single format called a message. This message format is used for all types of client queries and server responses, error messages, and the transfer of resource record information between servers.

Select each tab for more information.
Step 1
Step 2
Step 3
Step 4
Step 5

Step 1

The user types an FQDN into a browser application Address field.

this is a figure with a client contacting a DNS sever thru the network with a FQDN typed in a browser URL field because the name of a website is easier for people to used



The name is easy for people to use.
DNS Server
Network
Client

DNS Message Format

The DNS server stores different types of resource records that are used to resolve names. These records contain the name, address, and type of record. Some of these record types are as follows:

- **A** - An end device IPv4 address
- **NS** - An authoritative name server
- **AAAA** - An end device IPv6 address (pronounced quad-A)
- **MX** - A mail exchange record

When a client makes a query, the server DNS process first looks at its own records to resolve the name. If it is unable to resolve the name by using its stored records, it contacts other servers to resolve the name. After a match is found and returned to the original requesting server, the server temporarily stores the numbered address in the event that the same name is requested again.

The DNS cient service on Windows PCs also stores previously resolved names in memory. The **ipconfig /displaydns** command displays all of the cached DNS entries.

As shown in the table, DNS uses the same message format between servers, consisting of a question, answer, authority, and additional information for all types of client queries and server responses, error messages, and transfer of resource record information.

DNS message section	Description
Question	The question for the name server
Answer	Resource Records answering the question

Authority	Resource Records pointing toward an authorit
Additional	Resource Records holding additional informat

DNS Hierarchy

The DNS protocol uses a hierarchical system to create a database to provide name resolution, as shown in the figure. DNS uses domain names to form the hierarchy.

The naming structure is broken down into small, manageable zones. Each DNS server maintains a specific database file and is only responsible for managing name-to-IP mappings for that small portion of the entire DNS structure. When a DNS server receives a request for a name translation that is not within its DNS zone, the DNS server forwards the request to another DNS server within the proper zone for translation. DNS is scalable because hostname resolution is spread across multiple servers.

The different top-level domains represent either the type of organization or the country of origin. Examples of top-level domains are the following:

- **.com** - a business or industry
- **.org** - a non-profit organization
- **.au** - Australia
- **.co** - Colombia

The figure shows the DNS Hierachy tree. At the top is the Root Level Domain with the TOp-Level Domains(TLD) connected undererneath the Root Level Domainmain. THe TLDs are .net, .edu, .com,.au, .co, and other top-level doamins. Under the .com TLD is the Second Level domain www.cisco.com and under cisco.com are www.cisco.com , ftp.cisco.com, and mail.cisco.com.

Root Level Domain
Top-Level Domain (TLD)
Second Level Domain
cisco.com
www.cisco.com
ftp.cisco.com
mail.cisco.com
.net
.edu
.com
.au
.co
Other
Top-Level
Domains

The nslookup Command

When configuring a network device, one or more DNS Server addresses are provided that the DNS client can use for name resolution. Usually the ISP provides the addresses to use for the DNS servers. When a user application requests to connect to a remote device by name, the requesting DNS client queries the name server to resolve the name to a numeric address.

Computer operating systems also have a utility called Nslookup that allows the user to manually query the name servers to resolve a given host name. This utility can also be used to troubleshoot name resolution issues and to verify the current status of the name servers.

In this figure, when the **nslookup** command is issued, the default DNS server configured for your host is displayed. The name of a host or domain can be entered at the **nslookup** prompt. The Nslookup utility has many options available for extensive testing and verification of the DNS process.

```
C:\Users> nslookup
Default Server:  dns-sj.cisco.com
Address:  171.70.168.183
> www.cisco.com
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    origin-www.cisco.com
Addresses:  2001:420:1101:1::a
          173.37.145.84
Aliases:  www.cisco.com
> cisco.netacad.net
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    cisco.netacad.net
Address:  72.163.6.223
>
```

Dynamic Host Configuration Protocol

The Dynamic Host Configuration Protocol (DHCP) for IPv4 service automates the assignment of IPv4 addresses, subnet masks, gateways, and other IPv4 networking parameters. This is referred to as dynamic addressing. The alternative to dynamic addressing is static addressing. When using static addressing, the network administrator manually enters IP address information on hosts.

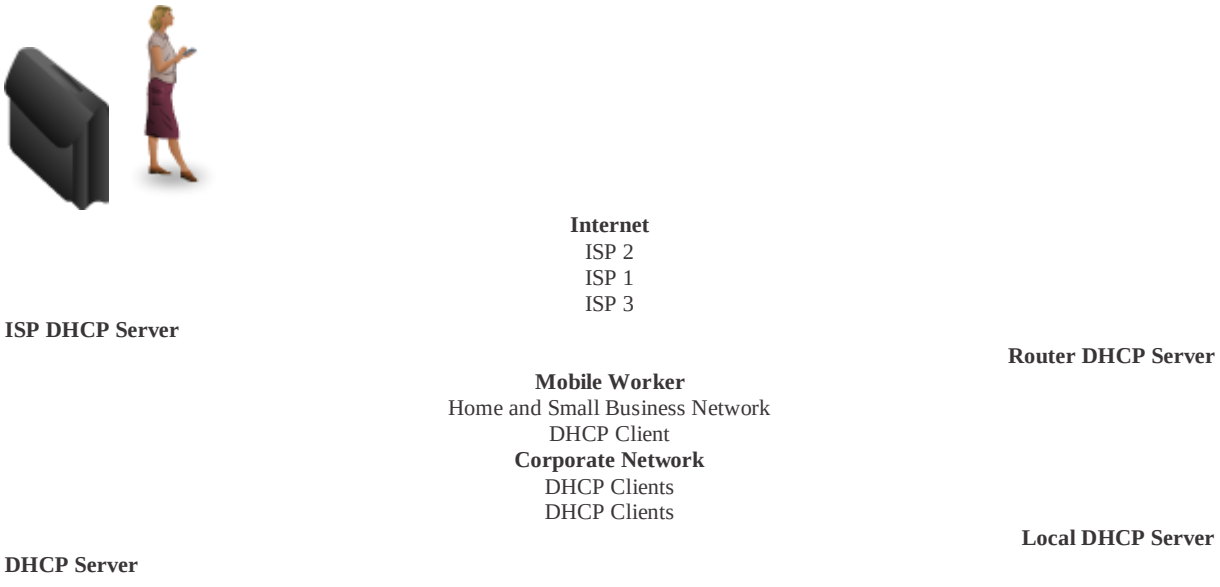
When a host connects to the network, the DHCP server is contacted, and an address is requested. The DHCP server chooses an address from a configured range of addresses called a pool and assigns (leases) it to the host.

On larger networks, or where the user population changes frequently, DHCP is preferred for address assignment. New users may arrive and need connections; others may have new computers that must be connected. Rather than use static addressing for each connection, it is more efficient to have IPv4 addresses assigned automatically using DHCP.

DHCP can allocate IP addresses for a configurable period of time, called a lease period. The lease period is an important DHCP setting, When the lease period expires or the DHCP server gets a DHCPRELEASE message the address is returned to the DHCP pool for reuse. Users can freely move from location to location and easily re-establish network connections through DHCP.

As the figure shows, various types of devices can be DHCP servers. The DHCP server in most medium-to-large networks is usually a local, dedicated PC-based server. With home networks, the DHCP server is usually located on the local router that connects the home network to the ISP.

The figure depicts a ISP DHCP server connected to the Internet with three ISP routers labelled ISP1, ISP2, ISP#. Each ISP reouter is connected to a different network. ISP1 connects to a wireless antenna to a mobile worker who is the DCHP client. ISP2 is connected to a coporate network router which connects to a coporate LAN with its own local DHCP server connected to a swith connected to six DHCP clients. ISP3 is connected to a wireless DHCP server for a Home and Small Business network the three DHCP clients connected.



Many networks use both DHCP and static addressing. DHCP is used for general purpose hosts, such as end user devices. Static addressing is used for network devices, such as gateway routers, switches, servers, and printers.

DHCP for IPv6 (DHCPv6) provides similar services for IPv6 clients. One important difference is that DHCPv6 does not provide a default gateway address. This can only be obtained dynamically from the Router Advertisement message of the router.

DHCP Operation

As shown in the figure, when an IPv4, DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP discover (DHCPDISCOVER) message to identify any available DHCP servers on the network. A DHCP server replies with a DHCP offer (DHCPOFFER) message, which offers a lease to the client. The offer message contains the IPv4 address and subnet mask to be assigned, the IPv4 address of the DNS server, and the IPv4 address of the default gateway. The lease offer also includes the duration of the lease.

The figure shows a protocol ladder with a DHCP client on one side and a DHCP client on the other. The DHCP client sends a DHCPDISCOVER message to the DHCP Server. The DHCP server sends a DHCPOFFER message to the DHCP client. The DHCP client sends a DHCPREQUEST message in repoonse to the DHCPOFFER from the DHCP server. THE DHCP server sends a DHCPACK message back to the DHCP client. The process is called DORA.

DHCP Client	DHCP Server
DHCPDISCOVER	
	DHCPOFFER
DHCPREQUEST	
	DHCPACK

The client may receive multiple DHCPOFFER messages if there is more than one DHCP server on the local network. Therefore, it must choose between them, and sends a DHCP request (DHCPREQUEST) message that identifies the explicit server and lease offer that the client is accepting. A client may also choose to request an address that it had previously been allocated by the server.

Assuming that the IPv4 address requested by the client, or offered by the server, is still available, the server returns a DHCP acknowledgment (DHCPACK) message that acknowledges to the client that the lease has been finalized. If the offer is no longer valid, then the selected server responds with a DHCP negative acknowledgment (DHCPNAK) message. If a DHCPNAK message is returned, then the selection process must begin again with a new DHCPDISCOVER message being transmitted. After the client has the lease, it must be renewed prior to the lease expiration through another DHCPREQUEST message.

The DHCP server ensures that all IP addresses are unique (the same IP address cannot be assigned to two different network devices simultaneously). Most ISPs use DHCP to allocate addresses to their customers.

DHCPv6 has a set of messages that is similar to those for DHCPv4. The DHCPv6 messages are SOLICIT, ADVERTISE, INFORMATION REQUEST, and REPLY.

5.1.5 File Sharing Services

File Transfer Protocol

As you learned in previous topics, in the client/server model, the client can upload data to a server, and download data from a server, if both devices are using a file transfer protocol (FTP). Like HTTP, email, and addressing protocols, FTP is commonly used application layer protocol. This topic discusses FTP in more detail.

FTP was developed to allow for data transfers between a client and a server. An FTP client is an application which runs on a computer that is being used to push and pull data from an FTP server.

The figure depicts an FTP transaction between a client and a server. A client is contacting a server thru a network. The first message from the client is a control connection: the client opens first connection to the server for control traffic. The second message from the client is a data connection: the client opens a second connection for data traffic. Data can then be downloaded from the server of uploaded from the client.

Server
Client
Network

1. Control Connection:
Client opens first connection to the server for control traffic.
2. Data Connection:
Client opens second connection for data traffic.

Get Data

3. Data Transfer:
Server transfers data to the client.

Based on commands sent across the control connection, data can be downloaded from the server or uploaded from the client.

The client establishes the first connection to the server for control traffic using TCP port 21. The traffic consists of client commands and server replies.

The client establishes the second connection to the server for the actual data transfer using TCP port 20. This connection is created every time there is data to be transferred.

The data transfer can happen in either direction. The client can download (pull) data from the server, or the client can upload (push) data to the server.

Server Message Block

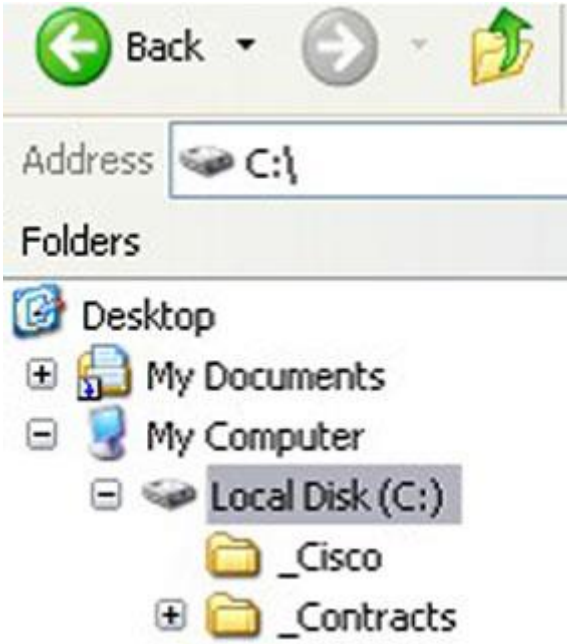
The Server Message Block (SMB) is a client/server file sharing protocol that describes the structure of shared network resources, such as directories, files, printers, and serial ports. It is a request-response protocol. All SMB messages share a common format. This format uses a fixed-sized header, followed by a variable-sized parameter and data component.

Here are three functions of SMB messages:

- Start, authenticate, and terminate sessions.
- Control file and printer access.
- Allow an application to send or receive messages to or from another device.

SMB file-sharing and print services have become the mainstay of Microsoft networking. With the introduction of the Windows 2000 software series, Microsoft changed the underlying structure for using SMB. In previous versions of Microsoft products, the SMB services used a non-TCP/IP protocol to implement name resolution. Beginning with Windows 2000, all subsequent Microsoft products use DNS naming, which allows TCP/IP protocols to directly support SMB resource sharing, as shown in the figure.

The first figure shows a Microsoft Windows shared resource of My Documents with a client requests from a Server My Documents. The client sends a SMB request and receives an SMB response of the shared resource My Documents. Shared Resources include File systems, Pronters shown as a icon, Mail slots, and APIs.



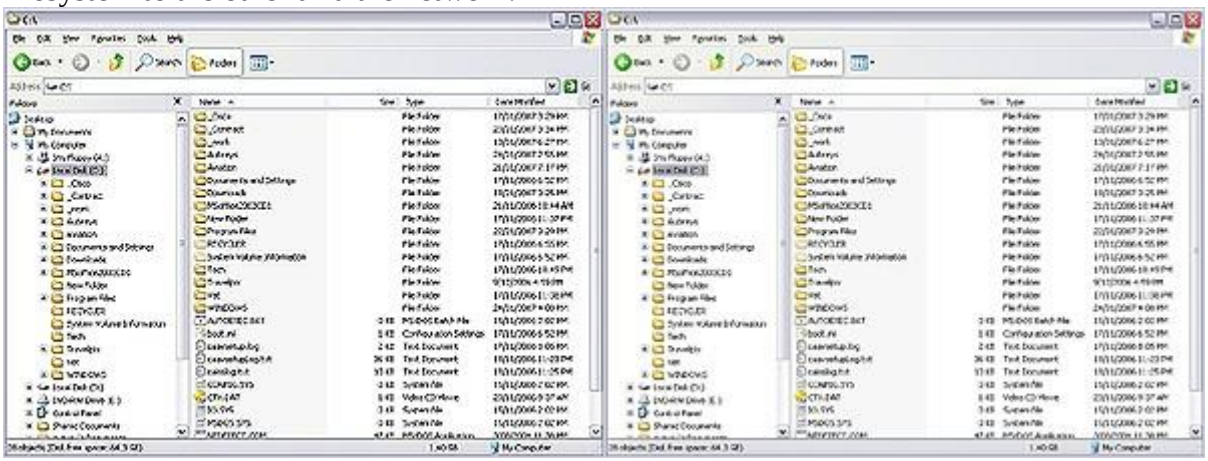
Client
SMB Requests
SMB Responses
Server
Printer
Shared Resources

- File systems
- Printers
- Mail slots
- APIs

SMB is a client/server, request-response protocol. Servers can make their own resources available to clients on the network.

The SMB file exchange process between Windows PCs is shown in the next figure.

The second figure depicts a copy file taking place between two Windows computers form one filesystem to the other thru the network.



Network
Copy File

A file may be copied from PC to PC with Windows Explorer using the SMB protocol.

Unlike the file sharing supported by FTP, clients establish a long-term connection to servers. After the connection is established, the user of the client can access the resources on the server as though the resource is local to the client host.

The LINUX and UNIX operating systems also provide a method of sharing resources with Microsoft networks by using a version of SMB called SAMBA. The Apple Macintosh operating systems also support resource sharing by using the SMB protocol.

5.3 Network Security Fundamentals

You may have already set up a network, or you may be getting ready to do just that. Here is something to think about. Setting up a network without securing it is like opening all the doors and windows to your home and then going on vacation. Anyone could come by, gain entry, steal or break items, or just make a mess. As you have seen on the news, it is possible to break into *any* network! As a network administrator, it is part of your job to make it difficult for threat actors to gain access to your network. This module gives you an overview of types of network attacks and what you can do to reduce a threat actor's chances of succeeding

5.3.1 Types of Threats

Types of Threats

Wired and wireless computer networks are essential to everyday activities. Individuals and organizations depend on their computers and networks. Intrusion by an unauthorized person can result in costly network outages and loss of work. Attacks on a network can be devastating and can result in a loss of time and money due to damage, or theft of important information or assets.

Intruders can gain access to a network through software vulnerabilities, hardware attacks, or through guessing someone's username and password. Intruders who gain access by modifying software or exploiting software vulnerabilities are called threat actors.

Information theft is breaking into a computer to obtain confidential information. Information can be used or sold for various purposes such as when someone is stealing proprietary information of an organization, like research and development data.

The figure shows an open folder

Types of Vulnerabilities

Vulnerability is the degree of weakness in a network or a device. Some degree of vulnerability is inherent in routers, switches, desktops, servers, and even security devices. Typically, the network devices under attack are the endpoints, such as servers and desktop computers.

There are three primary vulnerabilities or weaknesses: technological, configuration, and security policy. All three of these sources of vulnerabilities can leave a network or device open to various attacks, including malicious code attacks and network attacks.

Click each button for a table with examples and a description of each type of vulnerability.
Technological Vulnerabilities
Configuration Vulnerabilities
Policy Vulnerabilities

Vulnerability	Description
TCP/IP Protocol Weakness	• Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), and Internet Control Message Protocol (ICMP) are inherently insecure. • Simple Network Management Protocol (SNMP) and Simple Mail Transfer Protocol (SMTP) are related to the inherently insecure structure upon which TCP was designed.
Operating System Weakness	• Each operating system has security problems what must be addressed. • UNIX, Linux, Mac OS, Mac OS X, Windows Server 2012, Windows 7, Windows 8 • They are documented in the Computer Emergency Response Team (CERT) archives at http://www.cert.org
Network Equipment Weakness	Various types of network equipment, such as routers, firewalls, and switches have security weaknesses that must be recognized and protected against. Their weaknesses include password protection, lack of authentication, routing protocols, and firewall holes.

Physical Security

An equally important vulnerable area of the network to consider is the physical security of devices. If network resources can be physically compromised, a threat actor can deny the use of network resources.

The four classes of physical threats are as follows:

- **Hardware threats** - This includes physical damage to servers, routers, switches, cabling plant, and workstations.
- **Environmental threats** - This includes temperature extremes (too hot or too cold) or humidity extremes (too wet or too dry).
- **Electrical threats** - This includes voltage spikes, insufficient supply voltage (brownouts), unconditioned power (noise), and total power loss.
- **Maintenance threats** - This includes poor handling of key electrical components (electrostatic discharge), lack of critical spare parts, poor cabling, and poor labeling.

A good plan for physical security must be created and implemented to address these issues. The figure shows an example of physical security plan.

5.3.2 Network Attacks

Types of Malware

The previous topic explained the types of network threats and the vulnerabilities that make threats possible. This topic goes into more detail about how threat actors gain access to network or restrict authorized users from having access.

Malware is short for malicious software. It is code or software specifically designed to damage, disrupt, steal, or inflict “bad” or illegitimate action on data, hosts, or networks. Viruses, worms, and Trojan horses are types of malware.

Viruses

A computer virus is a type of malware that propagates by inserting a copy of itself into, and becoming part of, another program. It spreads from one computer to another, leaving infections as it travels. Viruses can range in severity from causing mildly annoying effects, to damaging data or software and causing denial of service (DoS) conditions. Almost all viruses are attached to an executable file, which means the virus may exist on a system but will not be active or able to spread until a user runs or opens the malicious host file or program. When the host code is executed, the viral code is executed as well. Normally, the host program keeps functioning after the virus infects it. However, some viruses overwrite other programs with copies of themselves, which destroys the host program altogether. Viruses spread when the software or document they are attached to is transferred from one computer to another using the network, a disk, file sharing, or infected email attachments.

Worms

Computer worms are similar to viruses in that they replicate functional copies of themselves and can cause the same type of damage. In contrast to viruses, which require the spreading of an infected host file, worms are standalone software and do not require a host program or human help to propagate. A worm does not need to attach to a program to infect a host and enter a computer through a vulnerability in the system. Worms take advantage of system features to travel through the network unaided.

Trojan Horses

A Trojan horse is another type of malware named after the wooden horse the Greeks used to infiltrate Troy. It is a harmful piece of software that looks legitimate. Users are typically tricked into loading and executing it on their systems. After it is activated, it can achieve any number of attacks on the host, from irritating the user (with excessive pop-up windows or changing the desktop) to damaging the host (deleting files, stealing data, or activating and spreading other malware, such as viruses). Trojan horses are also known to create back doors to give malicious users access to the system.

Unlike viruses and worms, Trojan horses do not reproduce by infecting other files. Trojan horses must spread through user interaction such as opening an email attachment or downloading and running a file from the internet.

Reconnaissance Attacks

In addition to malicious code attacks, it is also possible for networks to fall prey to various network attacks. Network attacks can be classified into three major categories:

- **Reconnaissance attacks**- The discovery and mapping of systems, services, or vulnerabilities.
- **Access attacks**- The unauthorized manipulation of data, system access, or user privileges.
- **Denial of service**- The disabling or corruption of networks, systems, or services.

For reconnaissance attacks, external threat actors can use internet tools, such as the **nslookup** and **whois** utilities, to easily determine the IP address space assigned to a given corporation or entity. After the IP address space is determined, a threat actor can then ping the publicly available IP addresses to identify the addresses that are active. To help automate this step, a threat actor may use a ping sweep tool, such as **fping** or **gping**. This systematically pings all network addresses in a given range or subnet. This is similar to going through a section of a telephone book and calling each number to see who answers.

Internet Queries

Click Play in the figure to view an animation. The threat actor is looking for initial information about a target. Various tools can be used, including Google search, the websites of organizations, whois, and more.

Threat Actor

Access Attacks

Access attacks exploit known vulnerabilities in authentication services, FTP services, and web services to gain entry to web accounts, confidential databases, and other sensitive information. An access attack allows individuals to gain unauthorized access to information that they have no right to view. Access attacks can be classified into four types: password attacks, trust exploitation, port redirection, and man-in-the middle.

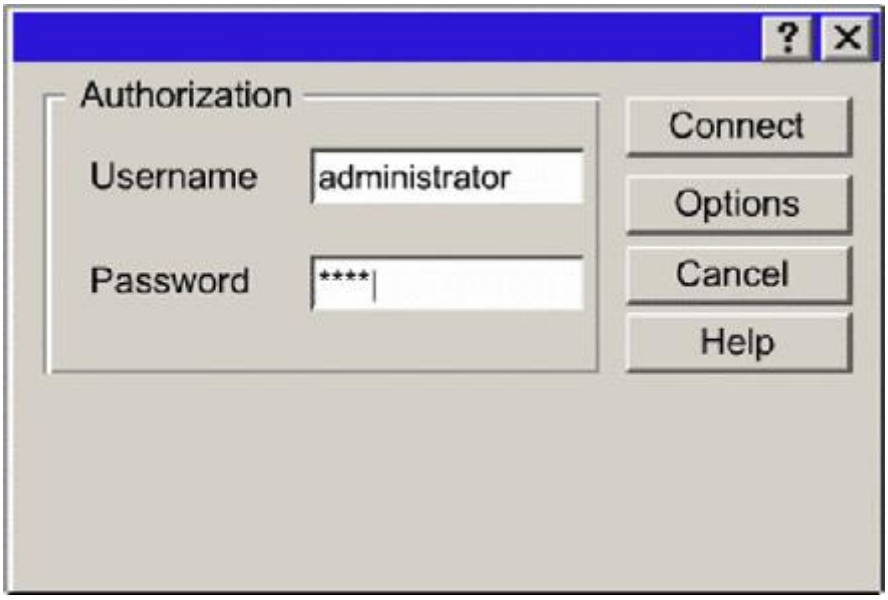
Click each button for an explanation of each type of attack.
Password Attacks
Trust Exploitation
Port Redirection
Man-in-the-Middle

Password Attacks

Threat actors can implement password attacks using several different methods:

- Brute-force attacks
- Trojan horse attacks
- Packet sniffers

The first figure shows a login prompt box with the username, administrator and the password, ****.



Denial of Service Attacks

Denial of service (DoS) attacks are the most publicized form of attack and among the most difficult to eliminate. However, because of their ease of implementation and potentially significant damage, DoS attacks deserve special attention from security administrators.

DoS attacks take many forms. Ultimately, they prevent authorized people from using a service by consuming system resources. To help prevent DoS attacks it is important to stay up to date with the latest security updates for operating systems and applications.

Click each button for an example of DoS and distributed DoS (DDoS) attacks.
DoS Attack
DDoS Attack

DoS Attack

DoS attacks are a major risk because they interrupt communication and cause significant loss of time and money. These attacks are relatively simple to conduct, even by an unskilled threat actor.

5.3.3 Network Attack Mitigations

The Defense-in-Depth Approach

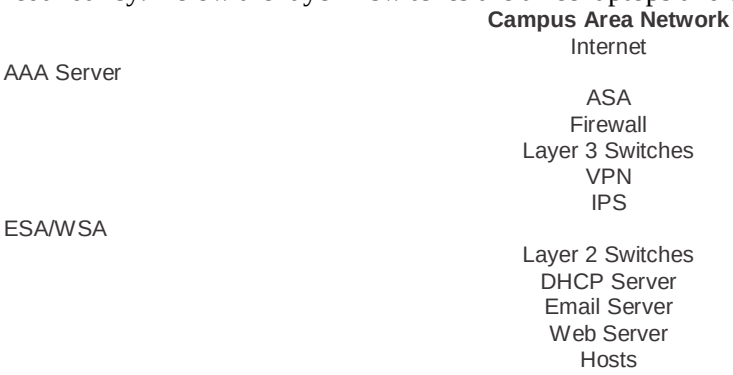
Now that you know more about how threat actors can break into networks, you need to understand what to do to prevent this unauthorized access. This topic details several actions you can take to make your network more secure.

To mitigate network attacks, you must first secure devices including routers, switches, servers, and hosts. Most organizations employ a defense-in-depth approach (also known as a layered approach) to security. This requires a combination of networking devices and services working in tandem.

Consider the network in the figure. There are several security devices and services that have been implemented to protect its users and assets against TCP/IP threats.

All network devices including the router and switches are also hardened as indicated by the combination locks on their respective icons. This indicates that they have been secured to prevent threat actors from gaining access and tampering with the devices.

The figure depicts a campus area network. A cloud representing the Internet is connected to a router, labeled VPN. The VPN router is connected to an ASA firewall. The firewall has two additional connections; one to an IPS and another to a switch. The switch is connected to a DHCP server, email server, web server, and ESA/WSA. The IPS is connected to a multilayer switch. The multilayer switch has a connection to an AAA server as well as to two layer 2 switches and a to another multilayer switch. The second multilayer switch also has connections to the same layer 2 switches, creating redundancy. Below the layer 2 switches are three laptops and three pcs which are labeled as hosts.



Several security devices and services are implemented to protect an organization's users and assets against TCP/IP threats.

- ***VPN*** *- A router is used to provide secure VPN services with corporate sites and remote access support for remote users using secure encrypted tunnels.*
- ***ASA Firewall*** *- This dedicated device provides stateful firewall services. It ensures that internal traffic can go out and come back, but external traffic cannot initiate connections to inside hosts.*
- ***IPS*** *- An intrusion prevention system (IPS) monitors incoming and outgoing traffic looking for malware, network attack signatures, and more. If it recognizes a threat, it can immediately stop it.*
- ***ESA/WSA*** *- The email security appliance (ESA) filters spam and suspicious emails. The web security appliance (WSA) filters known and suspicious internet malware sites.*
- ***AAA Server*** *- This server contains a secure database of who is authorized to access and manage network devices. Network devices authenticate administrative users using this database.*

Keep Backups

Backing up device configurations and data is one of the most effective ways of protecting against data loss. A data backup stores a copy of the information on a computer to removable backup media that can be kept in a safe place. Infrastructure devices should have backups of configuration files and IOS images on an FTP or similar file server. If the computer or a router hardware fails, the data or configuration can be restored using the backup copy.

Backups should be performed on a regular basis as identified in the security policy. Data backups are usually stored offsite to protect the backup media if anything happens to the main facility. Windows hosts have a backup and restore utility. It is important for users to back up their data to another drive, or to a cloud-based storage provider.

The table shows backup considerations and their descriptions.

Consideration	Description

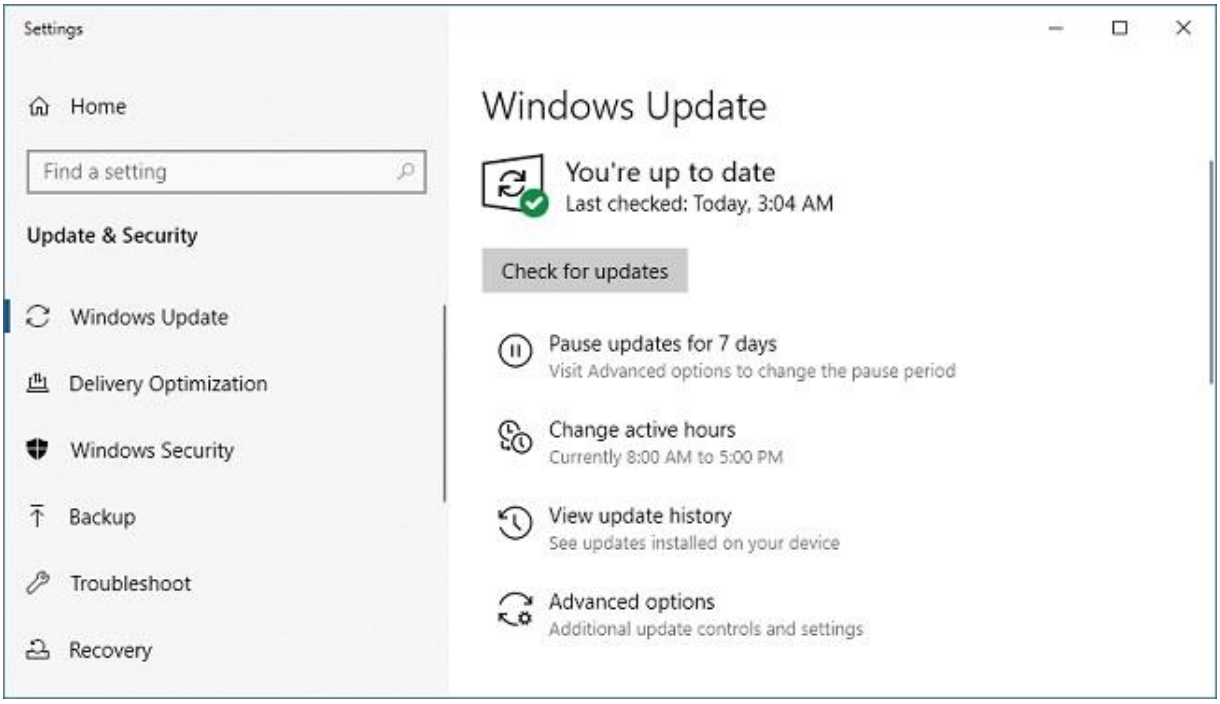
Frequency	• Perform backups on a regular basis as identified in the security policy. • Full backups can be time-consuming, therefore perform monthly or weekly backups with frequent partial backups of changed files.
Validation	• Always validate backups to ensure the integrity of the data and validate the file restoration procedures.
Storage	• Backups should be transported to an approved offsite storage location on a daily, weekly, or monthly rotation, as required by the security policy.
Security	• Backups should be protected using strong passwords. The password is required to restore the data.

Upgrade, Update, and Patch

Keeping up to date with the latest developments can lead to a more effective defense against network attacks. As new malware is released, enterprises need to keep current with the latest versions of antivirus software.

The most effective way to mitigate a worm attack is to download security updates from the operating system vendor and patch all vulnerable systems. Administering numerous systems involves the creation of a standard software image (operating system and accredited applications that are authorized for use on client systems) that is deployed on new or upgraded systems. However, security requirements change, and already deployed systems may need to have updated security patches installed.

One solution to the management of critical security patches is to make sure all end systems automatically download updates, as shown for Windows 10 in the figure. Security patches are automatically downloaded and installed without user intervention.



Authentication, Authorization, and Accounting

All network devices should be securely configured to provide only authorized individuals with access. Authentication, authorization, and accounting (AAA, or “triple A”) network security services provide the primary framework to set up access control on network devices.

AAA is a way to control who is permitted to access a network (authenticate), what actions they perform while accessing the network (authorize), and making a record of what was done while they are there (accounting).

The concept of AAA is similar to the use of a credit card. The credit card identifies who can use it, how much that user can spend, and keeps account of what items the user spent money on, as shown in the figure.

The figure shows a credit card next to a credit card statement. There is a rectangle around the numbers on the credit card with the text, Authentication Who are you? A second rectangle is around the credit limit on the credit card statement with text, Authorization How much can you spend? A third rectangle is around the transaction portion of the credit card summary with the text Accounting What did you spend on it?

Account Number	Statement Closing Date	Current Amount Due
1234-567-890	01-31-01	$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99900-1234

MAIL PAYMENT TO :
THE BANK
132 VINE STREET
ANYTOWN, USA 67500-0010

872919345 00178255000000003

Detach here and return upper portion with check or money order. Do not staple or fold.

Statement of Personal Credit Card Account

Retain this portion for your files.

THE BANK

Cardmember Name	Account Number	Statement Closing Date
JOE EMPLOYEE	1234-456-890	01-31-01

Statement Date:	02-01-01	Payment Due Date:	03-01-01
Closing Date:	01-31-01		
Credit Limit	$1,500.00	Credit Available:	$1221.50
New Balance:	$278.50	Minimum Payment Due:	$20.00

Account Summary

Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	$278.50

Reference Number	Sold	Posted	Activity Since Last Statement		Amount
43210987	01-03	01-13	Payment, Thank You		-$74.25
01234567	01-12	01-13	Wings 'N' Things	Anytown, USA	$25.25
78901234	01-14	01-17	Record Release	Anytown, USA	$40.00
45678901	01-14	01-17	Sports Stadium	Anytown, USA	$75.25
3210987	01-22	01-23	Tie Tack	Anytown, USA	$20.75
76543210	01-29	01-30	Electronic World	Anytown, USA	$89.25
23455678		01-30	Transaction Fees		$3.00
34567890		01-01	Annual Fee		$25.00

PAGE 1 OF 1

Authentication

Who are you?

Authorization

How much can you spend?
Accounting

What did you spend it on?

16.3.5 Firewalls

A firewall is one of the most effective security tools available for protecting users from external threats. A firewall protects computers and networks by preventing undesirable traffic from entering internal networks.

Network firewalls reside between two or more networks, control the traffic between them, and help prevent unauthorized access. For example, the top topology in the figure illustrates how the firewall enables traffic from an internal network host to exit the network and return to the inside network. The bottom topology illustrates how traffic initiated by the outside network (i.e., the internet) is denied access to the internal network.

Firewall Operation

The figure shows a rectangle, labeled Inside. Inside the rectangle there is a pc. Outside of and to the right of the rectangle, there is a firewall. To the right of the firewall, there is a cloud labeled, Internet. There are two arrows, one signifying traffic leaving the pc going through the firewall and out to the Internet. The second arrow signifies the firewall permitting traffic from the Internet to the pc. The figure shows another rectangle, labeled Inside. Inside the rectangle there is a pc. Outside of and to the right of the rectangle, there is a firewall. To the right of the firewall, there is a cloud labeled, Internet. There is an arrow pointing from the Internet to the firewall with an X signifying that traffic is being denied from the Internet to the internal network.

Firewall permits traffic from users in the inside network to exit and return.

Inside
Internet
Firewall

Firewall denies outside traffic access to the inside network.

Inside
Internet
Firewall

A firewall could allow outside users controlled access to specific services. For example, servers accessible to outside users are usually located on a special network referred to as the demilitarized zone (DMZ), as shown in the figure. The DMZ enables a network administrator to apply specific policies for hosts connected to that network.

Firewall Topology with DMZ

The figure shows a rectangle, labeled Inside. Inside the rectangle there is a pc. Outside of and to the right of the rectangle, there is a firewall. To the right of the firewall, there is a cloud labeled Internet. Above the firewall, theres a DMZ server inside of a rectangle. There are twos arrows, one going from the pc through the firewall to the DMZ server and another going from the Internet through the firewall to the DMZ sever.

DMZ
Server
Inside
Internet

Types of Firewalls

Firewall products come packaged in various forms. These products use different techniques for determining what will be permitted or denied access to a network. They include the following:

- **Packet filtering -** Prevents or allows access based on IP or MAC addresses
- **Application filtering -** Prevents or allows access by specific application types based on port numbers
- **URL filtering -** Prevents or allows access to websites based on specific URLs or keywords
- **Stateful packet inspection (SPI) -** Incoming packets must be legitimate responses to requests from internal hosts. Unsolicited packets are blocked unless permitted specifically. SPI can also include the capability to recognize and filter out specific types of attacks, such as denial of service (DoS)

Endpoint Security

An endpoint, or host, is an individual computer system or device that acts as a network client. Common endpoints are laptops, desktops, servers, smartphones, and tablets. Securing endpoint devices is one of the most challenging jobs of a network administrator because it involves human nature. A company must have well-documented policies in place and employees must be aware of these rules. Employees need to be trained on proper use of the network. Policies often include the use of antivirus software and host intrusion prevention. More comprehensive endpoint security solutions rely on network access control.

5.3.4 Device Security

One area of networks that requires special attention to maintain security is the devices. You probably already have a password for your computer, smart phone, or tablet. Is it as strong as it could be? Are you using other tools to enhance the security of your devices? This topic tells you how.

The security settings are set to the default values when a new operating system is installed on a device. In most cases, this level of security is inadequate. For Cisco routers, the Cisco AutoSecure feature can be used to assist securing the system, as shown in the example.

```
Router# auto secure
                --- AutoSecure Configuration ---
*** AutoSecure configuration enhances the security of
the router but it will not make router absolutely secure
from all security attacks ***
```

In addition, there are some simple steps that should be taken that apply to most operating systems:

- Default usernames and passwords should be changed immediately.
- Access to system resources should be restricted to only the individuals that are authorized to use those resources.
- Any unnecessary services and applications should be turned off and uninstalled when possible.

Often, devices shipped from the manufacturer have been sitting in a warehouse for a period of time and do not have the most up-to-date patches installed. It is important to update any software and install any security patches prior to implementation.

Passwords

To protect network devices, it is important to use strong passwords. Here are standard guidelines to follow:

- Use a password length of at least eight characters, preferably 10 or more characters. A longer password is a more secure password.
- Make passwords complex. Include a mix of uppercase and lowercase letters, numbers, symbols, and spaces, if allowed.
- Avoid passwords based on repetition, common dictionary words, letter or number sequences, usernames, relative or pet names, biographical information, such as birthdates, ID numbers, ancestor names, or other easily identifiable pieces of information.
- Deliberately misspell a password. For example, Smith = Smyth = 5mYth or Security = 5ecur1ty.
- Change passwords often. If a password is unknowingly compromised, the window of opportunity for the threat actor to use the password is limited.
- Do not write passwords down and leave them in obvious places such as on the desk or monitor.

The tables show examples of strong and weak passwords.

Weak Passwords

Weak Password	Why it is Weak
secret	Simple dictionary password
smith	Maiden name of mother
toyota	Make of a car
bob1967	Name and birthday of the user
Blueleaf23	Simple words and numbers

Strong Passwords

Strong Password	Why it is Strong
b67n42d39c	Combines alphanumeric characters
12^h u4@1p7	Combines alphanumeric characters, symbols, and includes a space

On Cisco routers, leading spaces are ignored for passwords, but spaces after the first character are not. Therefore, one method to create a strong password is to use the space bar and create a phrase made of many words. This is called a passphrase. A passphrase is often easier to remember than a simple password. It is also longer and harder to guess.

Additional Password Security

Strong passwords are only useful if they are secret. There are several steps that can be taken to help ensure that passwords remain secret on a Cisco router and switch including these:

- Encrypting all plaintext passwords
- Setting a minimum acceptable password length
- Deterring brute-force password guessing attacks
- Disabling an inactive privileged EXEC mode access after a specified amount of time.

As shown in the sample configuration in the figure, the **service password-encryption** global configuration command prevents unauthorized individuals from viewing plaintext passwords in the configuration file. This command encrypts all plaintext passwords. Notice in the example, that the password “cisco” has been encrypted as “03095A0F034F”.

To ensure that all configured passwords are a minimum of a specified length, use the **security passwords min-length** *length* command in global configuration mode. In the figure, any new password configured would have to have a minimum length of eight characters.

Threat actors may use password cracking software to conduct a brute-force attack on a network device. This attack continuously attempts to guess the valid passwords until one works. Use the **login block-for # attempts # within #** global configuration command to deter this type of attack. In the figure for example, the **login block-for 120 attempts 3 within 60** command will block vty login attempts for 120 seconds if there are three failed login attempts within 60 seconds.

Network administrators can become distracted and accidently leave a privileged EXEC mode session open on a terminal. This could enable an internal threat actor access to change or erase the device configuration.

By default, Cisco routers will logout an EXEC session after 10 minutes of inactivity. However, you can reduce this setting using the **exec-timeout** *minutes seconds* line configuration command. This command can be applied online console, auxiliary, and vty lines. In the figure, we are telling the Cisco device to automatically disconnect an inactive user on a vty line after the user has been idle for 5 minutes and 30 seconds.

```
R1(config)# service password-encryption
R1(config)# security passwords min-length 8
R1(config)# login block-for 120 attempts 3 within 60
R1(config)# line vty 0 4
R1(config-line)# password cisco123
R1(config-line)# exec-timeout 5 30
R1(config-line)# transport input ssh
R1(config-line)# end
R1#
R1# show running-config | section line vty
line vty 0 4
password 7 094F471A1A0A
exec-timeout 5 30
login
transport input ssh
R1#
```

Enable SSH

Telnet simplifies remote device access, but it is not secure. Data contained within a Telnet packet is transmitted unencrypted. For this reason, it is highly recommended to enable Secure Shell (SSH) on devices for secure remote access.

It is possible to configure a Cisco device to support SSH using the following six steps:

Step 1. Configure a unique device hostname. A device must have a unique hostname other than the default.

Step 2. Configure the IP domain name. Configure the IP domain name of the network by using the global configuration mode command **ip domain name** *name*.

Step 3. Generate a key to encrypt SSH traffic. SSH encrypts traffic between source and destination. However, to do so, a unique authentication key must be generated by using the global configuration command **crypto key generate rsa general-keys modulus** *bits*. The modulus *bits* determines the size of the key and can be configured from 360 bits to 2048 bits. The larger the bit value, the more secure the key. However, larger bit values also take longer to encrypt and decrypt information. The minimum recommended modulus length is 1024 bits.

Step 4. Verify or create a local database entry. Create a local database username entry using the **username** global configuration command. In the example, the parameter **secret** is used so that the password will be encrypted using MD5.

Step 5. Authenticate against the local database. Use the **login local** line configuration command to authenticate the vty line against the local database.

Step 6. Enable vty inbound SSH sessions. By default, no input session is allowed on vty lines. You can specify multiple input protocols including Telnet and SSH using the **transport input {ssh | telnet}** command.

As shown in the example, router R1 is configured in the span.com domain. This information is used along with the bit value specified in the **crypto key generate rsa general-keys modulus** command to create an encryption key.

Next, a local database entry for a user named Bob is created. Finally, the vty lines are configured to authenticate against the local database and to only accept incoming SSH sessions.

```
Router# configure terminal
Router(config)# hostname R1
R1(config)# ip domain name span.com
R1(config)# crypto key generate rsa general-keys modulus 1024
The name for the keys will be: R1.span.com % The key modulus
size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-
exportable...[OK]
Dec 13 16:19:12.079: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#
R1(config)# username Bob secret cisco
R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
R1(config-line)# exit
R1(config)#
```

Disable Unused Services

Cisco routers and switches start with a list of active services that may or may not be required in your network. Disable any unused services to preserve system resources, such as CPU cycles and RAM, and prevent threat actors from exploiting these services. The type of services that are on by default will vary depending on the IOS version. For example, IOS-XE typically will have only HTTPS and DHCP ports open. You can verify this with the **show ip ports all** command, as shown in the example.

```
Router# show ip ports all
Proto Local Address                Foreign
Address              State        PID/Program Name
TCB        Local Address                Foreign
Address              (state)
tcp    :::443                      :::*                          LI
STEN       309/[IOS]HTTP CORE
tcp    *:443                       *:*                           LI
STEN       309/[IOS]HTTP CORE
udp    *:67                        0.0.0.0:0
            387/[IOS]DHCPD Receive
Router#
```

IOS versions prior to IOS-XE use the **show control-plane host open-ports** command. We mention this command because you may see it on older devices. The output is similar. However, notice that this older router has an insecure HTTP server and Telnet running. Both of these services should be disabled. As shown in the example, disable HTTP with the **no ip http server** global configuration command. Disable Telnet by specifying only SSH in the line configuration command, **transport input ssh**.

```
Router# show control-plane host open-ports
Active internet connections (servers and established)
Prot        Local Address      Foreign
Address                  Service    State
tcp                *:23                  *:0
Telnet   LISTEN
tcp                *:80                  *:0                  HT
TP CORE   LISTEN
udp                *:67                  *:0             DHCPD
Receive   LISTEN
Router# configure terminal
Router(config)# no ip http server
Router(config)# line vty 0 15
Router(config-line)# transport input ssh
```

5.4 Build A Small Network

5.4.1 Devices in Small Network

Small Network Topologies

The majority of businesses are small; therefore, it is not surprising that the majority of business networks are also small.

A small network design is usually simple. The number and type of devices included are significantly reduced compared to that of a larger network.

For instance, refer to the sample small-business network shown in the figure.

small network topology with a printer, server, IP phone and attached host, access point and attached laptop, all connected to a switch connected to a router connected to the Internet cloud
Internet

This small network requires a router, a switch, and a wireless access point to connect wired and wireless users, an IP phone, a printer, and a server. Small networks typically have a single WAN connection provided by DSL, cable, or an Ethernet connection.

Large networks require an IT department to maintain, secure, and troubleshoot network devices and to protect organizational data. Managing a small network requires many of the same skills as those required for managing a larger one. Small networks are managed by a local IT technician or by a contracted professional.

Device Selection for a Small Network

Like large networks, small networks require planning and design to meet user requirements. Planning ensures that all requirements, cost factors, and deployment options are given due consideration.

One of the first design considerations is the type of intermediary devices to use to support the network.

Click each button for more information about the factors that must be considered when selecting network devices.
Cost
Speed and Types of Ports/Interfaces
Expandability
Operating System Features and Services

Cost

The cost of a switch or router is determined by its capacity and features. This includes the number and types of ports available and the backplane speed. Other factors that influence the cost are network management capabilities, embedded security technologies, and optional advanced switching technologies. The expense of cable runs required to connect every device on the network must also be considered. Another key element affecting cost considerations is the amount of redundancy to incorporate into the network.

IP Addressing for a Small Network

When implementing a network, create an IP addressing scheme and use it. All hosts and devices within an internetwork must have a unique address.

Devices that will factor into the IP addressing scheme include the following:

- End user devices - The number and type of connection (i.e., wired, wireless, remote access)
- Servers and peripherals devices (e.g., printers and security cameras)
- Intermediary devices including switches and access points

It is recommended that you plan, document, and maintain an IP addressing scheme based on device type. The use of a planned IP addressing scheme makes it easier to identify a type of device and to troubleshoot problems, as for instance, when troubleshooting network traffic issues with a protocol analyzer.

For example, refer to the topology of a small to medium sized organization in the figure.

network topology consisting of three LANs - 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24 - with various end devices, connected to a router connected to the Internet cloud
192.168.1.0/24192.168.2.0/24192.168.3.0/24
Internet

The organization requires three user LANs (i.e., 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24). The organization has decided to implement a consistent IP addressing scheme for each 192.168.x.0/24 LAN using the following plan:

Device Type	**Assignable IP Address Range**	**Summarized as …**
Default gateway (Router)	192.168.x.**1** - 192.168.x.**2**	192.168.x.**0/30**
Switches (max 2)	192.168.x.**5** - 192.168.x.**6**	192.168.x.**4/30**
Access points (max 6)	192.168.x.**9** - 192.168.x.**14**	192.168.x.**8/29**
Servers (max 6)	192.168.x.**17** - 192.168.x.**22**	192.168.x.**16/29**
Printers (max 6)	192.168.x.**25** - 192.168.x.**30**	192.168.x.**24/29**
IP Phones (max 6)	192.168.x.**33** - 192.168.x.**38**	192.168.x.**32/29**
Wired devices (max 62)	192.168.x.**65** - 192.168.x.**126**	192.168.x.**64/26**
Wireless devices (max 62)	192.168.x.**193** - 192.168.x.**254**	192.168.x.**192/26**

The figure displays an example of the 192.168.2.0/24 network devices with assigned IP addresses using the predefined IP addressing scheme.

The diagram is a small LAN topology with a network address of 192.168.2.0/24. It shows various end devices all connected to a switch, with address .5, connected to a router, at address .1, connected to the Internet cloud. All devices have been assigned an IP address. A printer has an address of .25; server has an address of .17; a PC has an address of .65 connected to an IP phone with an address of .33; and a laptop has an address of .193 connected to an access point with an address of .9.
.193.9.5.33.65.25.17.1192.168.2.0/24
Internet

For instance, the default gateway IP address is 192.168.2.1/24, the switch is 192.168.2.5/24, the server is 192.168.2.17/24, etc..

Notice that the assignable IP address ranges were deliberately allocated on subnetnetwork boundaries to simplify summarizing the group type. For instance, assume another switch with IP address 192.168.2.6 is added to the network. To identify all switches in a network policy, the administrator could specify the summarized network address 192.168.x.4/30.

Redundancy in a Small Network

Another important part of network design is reliability. Even small businesses often rely heavily on their network for business operation. A failure of the network can be very costly.

In order to maintain a high degree of reliability, *redundancy* is required in the network design. Redundancy helps to eliminate single points of failure.

There are many ways to accomplish redundancy in a network. Redundancy can be accomplished by installing duplicate equipment, but it can also be accomplished by supplying duplicate network links for critical areas, as shown in the figure.

The diagram illustrates the use of redundant servers, links, switches, and routers in a network. Shown are four layers with an explanation of the redundancy achieved at each. The top layer has three servers and text reads: Redundant servers are available in case of server failure. The next layer shows that each server has two connections leading to two switches and text reads: Redundant links are present to provide alternate paths in case of a link failure. The next layer shows two switches connected to each other with each connected to all three servers above and text reads: Redundant switches are present in case of switch failure. The bottom layer shows two routers connected to each other with each connected to one of the switches and text reads: Redundant routers are available in case of router or route failure.
Redundant servers are available in case of server failure.
Redundant links are present to provide alternate paths in case of a link failure.
Redundant switches are present in case of switch failure.
Redundant routers are available in case of router or route failure.

Small networks typically provide a single exit point toward the internet via one or more default gateways. If the router fails, the entire network loses connectivity to the internet. For this reason, it may be advisable for a small business to pay for a second service provider as backup.

Traffic Management

The goal for a good network design, even for a small network, is to enhance the productivity of the employees and minimize network downtime. The network administrator should consider the various types of traffic and their treatment in the network design.

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5.4.2 Small Network Applications and Protocols

Common Applications

The previous topic discussed the components of a small network, as well as some of the design considerations. These considerations are necessary when you are just setting up a network. After you have set it up, your network still needs certain types of applications and protocols in order to work.

The network is only as useful as the applications that are on it. There are two forms of software programs or processes that provide access to the network: network applications and application layer services.

Network Applications

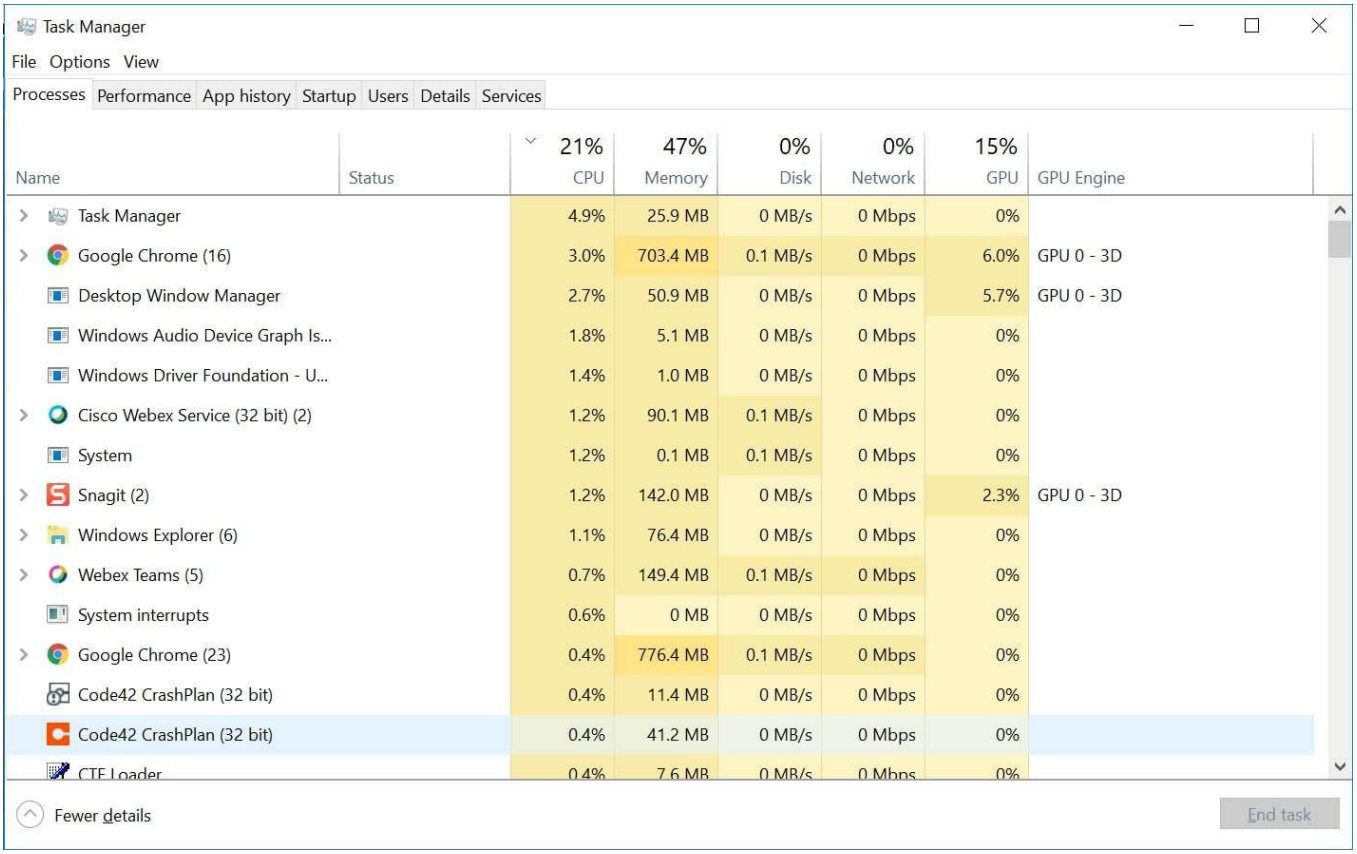
Applications are the software programs used to communicate over the network. Some end-user applications are network-aware, meaning that they implement application layer protocols and are able to communicate directly with the lower layers of the protocol stack. Email clients and web browsers are examples of this type of application.

Application Layer Services

Other programs may need the assistance of application layer services to use network resources like file transfer or network print spooling. Though transparent to an employee, these services are the programs that interface with the network and prepare the data for transfer. Different types of data, whether text, graphics or video, require different network services to ensure that they are properly prepared for processing by the functions occurring at the lower layers of the OSI model.

Each application or network service uses protocols, which define the standards and data formats to be used. Without protocols, the data network would not have a common way to format and direct data. In order to understand the function of various network services, it is necessary to become familiar with the underlying protocols that govern their operation.

Use the Task Manager to view the current applications, processes, and services running on a Windows PC, as shown in the figure.



Common Protocols

Most of a technician's work, in either a small or a large network, will in some way be involved with network protocols. Network protocols support the applications and services used by employees in a small network.

Network administrators commonly require access to network devices and servers. The two most common remote access solutions are Telnet and Secure Shell (SSH). SSH service is a secure alternative to Telnet. When connected, administrators can access the SSH server device as though they were logged in locally.

SSH is used to establish a secure remote access connection between an SSH client and other SSH-enabled devices:

- **Network device** - The network device (e.g., router, switch, access point, etc.) must support SSH to provide remote access SSH server services to clients.
- **Server** - The server (e.g., web server, email server, etc.) must support remote access SSH server services to clients.

Web Server

- Web clients and web servers exchange web traffic using the Hypertext Transfer Protocol (HTTP).
- Hypertext Transfer Protocol Secure (HTTPS) is used for secure web communication.

Note: A server could provide multiple network services. For instance, a server could be an email, FTP, and SSH server.

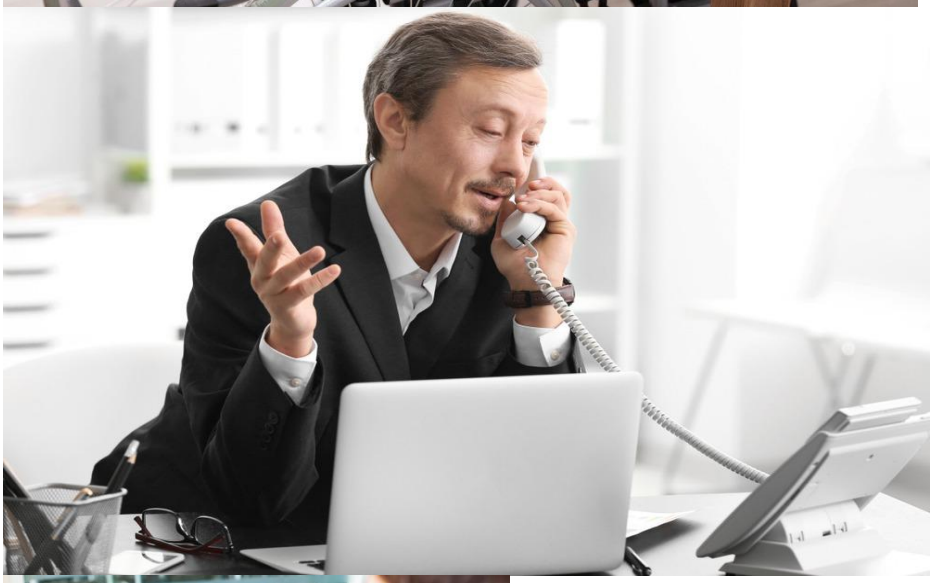
These network protocols comprise the fundamental toolset of a network professional. Each of these network protocols define:

- Processes on either end of a communication session
- Types of messages
- Syntax of the messages
- Meaning of informational fields
- How messages are sent and the expected response
- Interaction with the next lower layer

Many companies have established a policy of using secure versions (e.g., SSH, SFTP, and HTTPS) of these protocols whenever possible.

Voice and Video Applications

Businesses today are increasingly using IP telephony and streaming media to communicate with customers and business partners. Many organizations are enabling their employees to work remotely. As the figure shows, many of their users still require access to corporate software and files, as well as support for voice and video applications.



The network administrator must ensure the proper equipment is installed in the network and that the network devices are configured to ensure priority delivery.

Click each button for more information about the factors that a small network administrator must consider when supporting real-time applications.
Infrastructure
VoIP
IP Telephony

Real-Time Applications

Infrastructure

- The network infrastructure must support the real-time applications.
- Existing devices and cabling must be tested and validated.
- Newer networking products may be required.

5.4.3 Scale to Larger Networks

Small Network Growth

If your network is for a small business, presumably, you want that business to grow, and your network to grow along with it. This is called scaling a network, and there are some best practices for doing this.

Growth is a natural process for many small businesses, and their networks must grow accordingly. Ideally, the network administrator has enough lead-time to make intelligent decisions about growing the network in alignment with the growth of the company.

To scale a network, several elements are required:

- **Network documentation** - Physical and logical topology
- **Device inventory** - List of devices that use or comprise the network
- **Budget** - Itemized IT budget, including fiscal year equipment purchasing budget
- **Traffic analysis** - Protocols, applications, and services and their respective traffic requirements should be documented

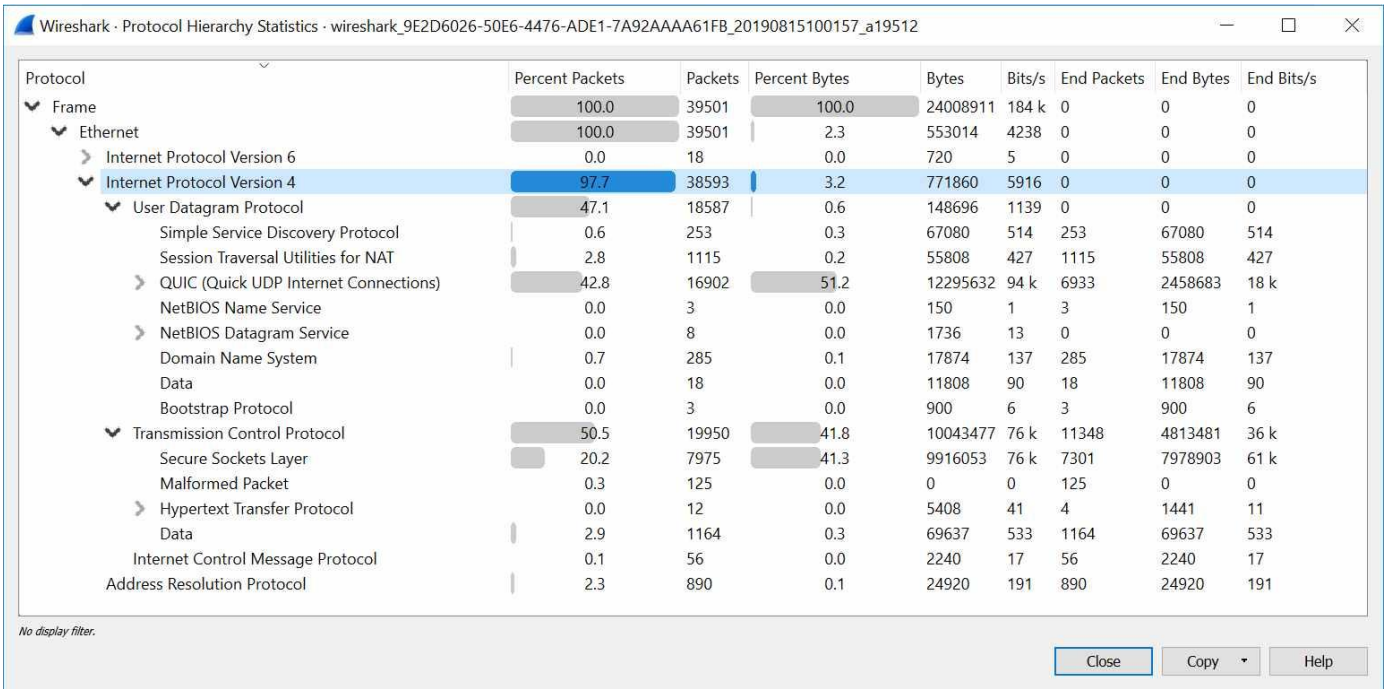
These elements are used to inform the decision-making that accompanies the scaling of a small network.

Protocol Analysis

As the network grows, it becomes important to determine how to manage network traffic. It is important to understand the type of traffic that is crossing the network as well as the current traffic flow. There are several network management tools that can be used for this purpose. However, a simple protocol analyzer such as Wireshark can also be used.

For instance, running Wireshark on several key hosts can reveal the types of network traffic flowing through the network. The following figure displays Wireshark protocol hierarchy statistics for a Windows host on a small network.

screen capture of Wireshark protocol hierarchy statistics for traffic captured by a host



Wireshark · Protocol Hierarchy Statistics · wireshark_9E2D6026-50E6-4476-ADE1-7A92AAAA61FB_20190815100157_a19512

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s
Frame	100.0	39501	100.0	24008911	184 k	0	0	0
Ethernet	100.0	39501	2.3	553014	4238	0	0	0
Internet Protocol Version 6	0.0	18	0.0	720	5	0	0	0
Internet Protocol Version 4	97.7	38593	3.2	771860	5916	0	0	0
User Datagram Protocol	47.1	18587	0.6	148696	1139	0	0	0
Simple Service Discovery Protocol	0.6	253	0.3	67080	514	253	67080	514
Session Traversal Utilities for NAT	2.8	1115	0.2	55808	427	1115	55808	427
QUIC (Quick UDP Internet Connections)	42.8	16902	51.2	12295632	94 k	6933	2458683	18 k
NetBIOS Name Service	0.0	3	0.0	150	1	3	150	1
NetBIOS Datagram Service	0.0	8	0.0	1736	13	0	0	0
Domain Name System	0.7	285	0.1	17874	137	285	17874	137
Data	0.0	18	0.0	11808	90	18	11808	90
Bootstrap Protocol	0.0	3	0.0	900	6	3	900	6
Transmission Control Protocol	50.5	19950	41.8	10043477	76 k	11348	4813481	36 k
Secure Sockets Layer	20.2	7975	41.3	9916053	76 k	7301	7978903	61 k
Malformed Packet	0.3	125	0.0	0	0	125	0	0
Hypertext Transfer Protocol	0.0	12	0.0	5408	41	4	1441	11
Data	2.9	1164	0.3	69637	533	1164	69637	533
Internet Control Message Protocol	0.1	56	0.0	2240	17	56	2240	17
Address Resolution Protocol	2.3	890	0.1	24920	191	890	24920	191

No display filter.

Close Copy Help

The screen capture reveals the host is using IPv6 and IPv4 protocols. The IPv4 specific output also reveals that the host has used DNS, SSL, HTTP, ICMP, and other protocols.

To determine traffic flow patterns, it is important to do the following:

- Capture traffic during peak utilization times to get a good representation of the different traffic types.
- Perform the capture on different network segments and devices as some traffic will be local to a particular segment.

Information gathered by the protocol analyzer is evaluated based on the source and destination of the traffic, as well as the type of traffic being sent. This analysis can be used to make decisions on how to manage the traffic more efficiently. This can be done by reducing unnecessary traffic flows or changing flow patterns altogether by moving a server, for example.

Sometimes, simply relocating a server or service to another network segment improves network performance and accommodates the growing traffic needs. At other times, optimizing the network performance requires major network redesign and intervention.

Employee Network Utilization

In addition to understanding changing traffic trends, a network administrator must be aware of how network use is changing. Many operating systems provide built-in tools to display such information. For example, a Windows host provides tools such as the Task Manager, Event Viewer, and Data Usage tools.

These tools can be used to capture a “snapshot” of information such as the following:

- OS and OS Version
- CPU utilization

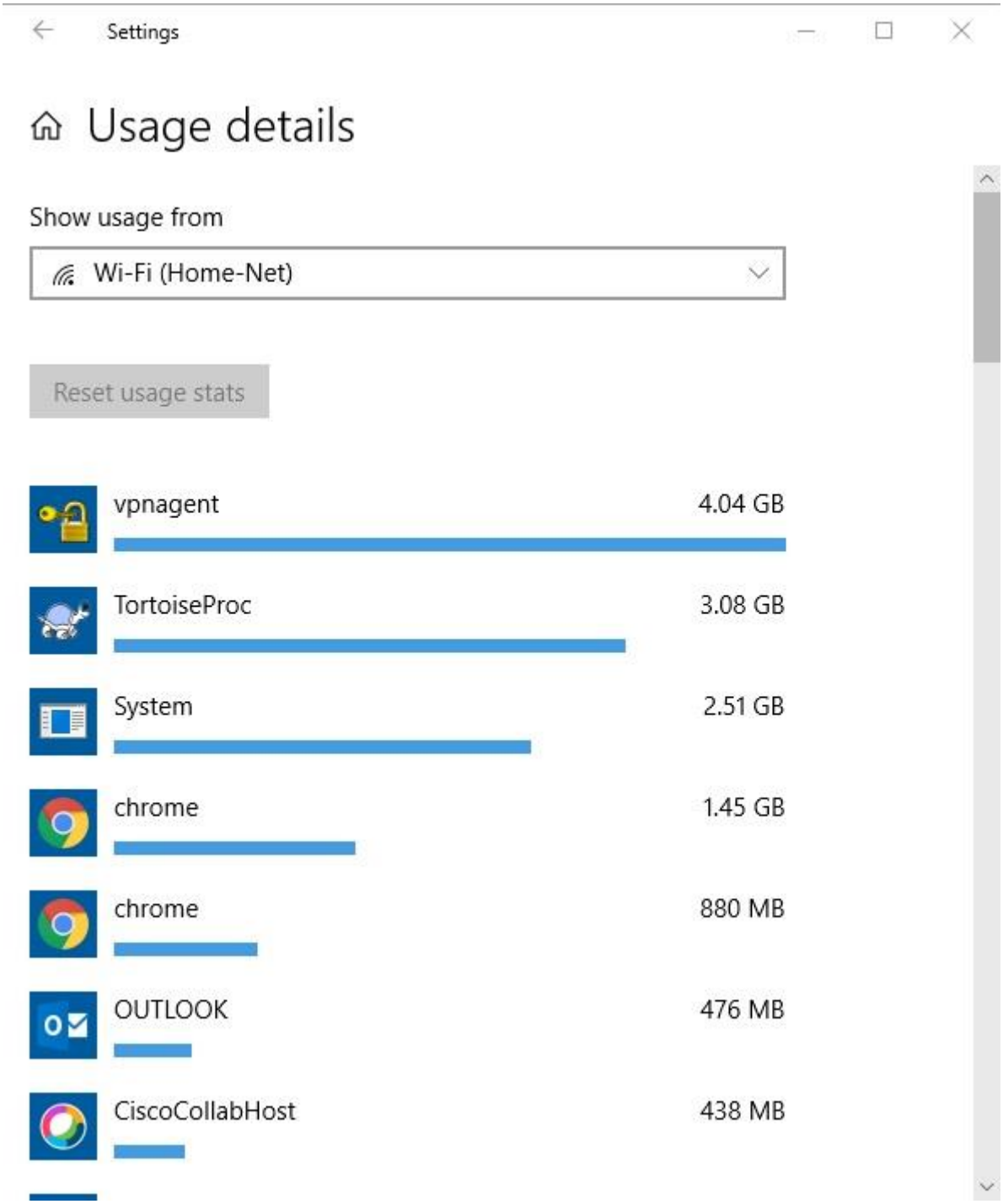
- RAM utilization
- Drive utilization
- Non-Network applications
- Network applications

Documenting snapshots for employees in a small network over a period of time is very useful to identify evolving protocol requirements and associated traffic flows. A shift in resource utilization may require the network administrator to adjust network resource allocations accordingly.

The Windows 10 Data Usage tool is especially useful to determine which applications are using network services on a host. The Data Usage tool is accessed using **Settings > Network & Internet > Data usage > network interface** (from the last 30 days).

The example in the figure is displaying the applications running on a remote user Windows 10 host using the local Wi-Fi network connection.

screen capture of the Windows 10 Data Usage Tool showing usage from a local Wi-Fi connection



5.4.4 Verify Connectivity

Verify Connectivity with Ping

Whether your network is small and new, or you are scaling an existing network, you will always want to be able to verify that your components are properly connected to each other and to the internet. This topic discusses some utilities that you can use to ensure that your network is connected.

The **ping** command is the most effective way to quickly test Layer 3 connectivity between a source and destination IP address. The command also displays various round-trip time statistics.

Specifically, the **ping** command uses the Internet Control Message Protocol (ICMP) echo request (ICMP Type 8) and echo reply (ICMP Type 0) messages. The **ping** command is available in most operating systems including Windows, Linux, macOS, and Cisco IOS.

On a Windows 10 host, the **ping** command sends four consecutive ICMP echo request messages and expects four consecutive ICMP echo replies from the destination.

For example, assume PC A pings PC B. As shown in the figure, the PC A Windows host sends four consecutive ICMP echo request messages. sometimes referred to as an ICMP echo, to PC B (i.e., 203.0.113.8).

The diagram shows host PC A, at address 192.168.10.10, using the ping ping 203.0.113.8 command from the command prompt to send four ICMP echo messages with a source IP of 198.168.10.10 (should read 192.168.10.10) and a destination IP of ping 203.0.113.8, which is host PC B on another network.
192.168.10.10PC APC BC:\>ping 203.0.113.8203.0.113.8

Source IP	Destination IP	ICMP
192.168.10.10	203.0.113.8	**Echo**

Internet

The destination host receives and processes the ICMP echos. As shown in the figure, PC B responds by sending four ICMP echo reply messages to PC A.

The diagram shows host PC B, at address 203.0.113.8, sending four ICMP echo replies with source IP 203.0.113.8 and destination IP 198.168.10.10 (should read 192.168.10.10) in response to a ping from host PC A at address 192.168.10.10.
PC APC BC:\>ping 203.0.113.8203.0.113.8192.168.10.10

SourceIP	Destination IP	ICMP
203.0.113.8	192.168.10.10	**Echo Replies**

Internet

As shown in the command output, PC A has received echo replies from PC-B verifying the Layer 3 network connection.

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time=47ms TTL=51
Reply from 10.1.1.10: bytes=32 time=60ms TTL=51
Reply from 10.1.1.10: bytes=32 time=53ms TTL=51
Reply from 10.1.1.10: bytes=32 time=50ms TTL=51
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
```

```
    Minimum = 47ms, Maximum = 60ms, Average = 52ms
C:\Users\PC-A>
```

The output validates Layer 3 connectivity between PC A and PC B.

A Cisco IOS **ping** command output varies from a Windows host. For instance, the IOS ping sends five ICMP echo messages, as shown in the output.

```
R1# ping 10.1.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.10, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
R1#
```

Notice the **!!!!!** output characters. The IOS **ping** command displays an indicator for each ICMP echo reply received. The table lists the most common output characters from the **ping** command.

IOS Ping Indicators

Element	Description
!	• Exclamation mark indicates successful receipt of an echo reply message. • It validates a Layer 3 connection between source and destination.
.	• A period means that time expired waiting for an echo reply message. • This indicates a connectivity problem occurred somewhere along the path.
U	• Uppercase **U** indicates a router along the path responded with an ICMP Type 3 “destination unreachable” error message. • Possible reasons include the router does not know the direction to the destination network or it could not find the host on the destination network.

Note: Other possible ping replies include Q, M, ?, or &. However, the meaning of these are out of scope for this module.

Extended Ping

A standard **ping** uses the IP address of the interface closest to the destination network as the source of the **ping**. The source IP address of the **ping 10.1.1.10** command on R1 would be that of the G0/0/0 interface (i.e., 209.165.200.225), as illustrated in the example.

The diagram shows how a router uses a standard ping to ping a host by sending four consecutive ICMP echo messages sourced from the interface closest to the destination. Router R1 is connected to two networks: on the left is 192.168.10.0/24 on interface G0/0/1 with address .1 and on the right is network 209.165.200.224/30 on interface G0/0/0 with address .225. The latter network is connected to R2 which is connected to network 10.1.1.0/24 on which host PC B is attached with address .10. R1 is sending PC B four ICMP echo messages with a source IP of 209.165.200.225 and a destination IP of 10.1.1.10.
R2.10.1G0/0/0.10209.165.200.224 /30192.168.10.0 /2410.1.1.0/24.225G0/0/1PC APC BR1

Source IP	Destination IP	ICMP
209.165.200.225	10.1.1.10	**Echo**

The Cisco IOS offers an "extended" mode of the **ping** command. This mode enables the user to create special type of pings by adjusting parameters related to the command operation.

Extended ping is entered in privileged EXEC mode by typing **ping** without a destination IP address. You will then be given several prompts to customize the extended **ping**.

Note: Pressing **Enter** accepts the indicated default values.

For example, assume you wanted to test connectivity from the R1 LAN (i.e., 192.168.10.0/24) to the 10.1.1.0 LAN. This could be verified from the PC A. However, an extended **ping** could be configured on R1 to specify a different source address.

As illustrated in the example, the source IP address of the extended **ping** command on R1 could be configured to use the G0/0/1 interface IP address (i.e., 192.168.10.1).

The diagram shows how a router uses an extended ping command to ping a host by sending four consecutive ICMP echo messages with a specified source IP address. Router R1 is connected to two networks: on the left is 192.168.10.0/24 on interface G0/0/1 with address .1 and on the right is network 209.165.200.224/30 on interface G0/0/0 with address .225. The latter network is connected to R2 which is connected to network 10.1.1.0/24 on which host PC B is attached with address .10. R1 is sending PC B four ICMP echo messages with a source IP of 192.168.10.1 and a destination IP of 10.1.1.10.
R2.10.1G0/0/0.10209.165.200.224 /30192.168.10.0 /2410.1.1.0/24.225G0/0/1PC APC BR1

Source IP	Destination IP	ICMP
192.168.10.1	10.1.1.10	**Echo**

The following command output configures an extended **ping** on R1 and specifies the source IP address to be that of the G0/0/1 interface (i.e., 192.168.10.1).

```
R1# ping
Protocol [ip]:
Target IP address: 10.1.1.10
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Ingress ping [n]:
Source address or interface: 192.168.10.1
DSCP Value [0]:
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0x0000ABCD]:
```

```
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2
seconds:
Packet sent with a source address of 192.168.10.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/1 ms
R1#
```

Note: The **ping ipv6** command is used for IPv6 extended pings.

Verify Connectivity with Traceroute

The **ping** command is useful to quickly determine if there is a Layer 3 connectivity problem. However, it does not identify where the problem is located along the path.

Traceroute can help locate Layer 3 problem areas in a network. A trace returns a list of hops as a packet is routed through a network. It could be used to identify the point along the path where the problem can be found.

The syntax of the trace command varies between operating systems, as illustrated in the figure.

Windows and Cisco IOS Trace Commands

The diagram shows the difference between the trace command as issued from a Windows host versus a Cisco IOS router. The network topology consists of a host PC A connected to a switch connected to router R1 connected to router R2 connected to router R3 connected to a switch connected to host PC B. PC A, at IP address 192.168.10.10, is issuing the following command from a Windows command prompt: C:\>:tracert 10.1.1.10. R1 is issuing the following command from the Cisco IOS CLI: R#traceroute 10.1.1.10.

PC APC B10.1.1.10192.168.10.10.1R3R2R1

Trace from Windows host
C:\>:**tracert 10.1.1.10**
Trace from a Cisco IOS router
R# **traceroute 10.1.1.10**

The following is a sample output of **tracert** command on a Windows 10 host.

```
C:\Users\PC-A> tracert 10.1.1.10
Tracing route to 10.1.10 over a maximum of 30 hops:
  1     2 ms     2 ms     2 ms  192.168.10.1
  2     *        *        *     Request timed out.
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
^C
C:\Users\PC-A>
```

Note: Use **Ctrl-C** to interrupt a **tracert** in Windows.

The only successful response was from the gateway on R1. Trace requests to the next hop timed out as indicated by the asterisk (*), meaning that the next hop router did not respond. The timed out requests indicate that there is a failure in the internetwork beyond the LAN, or that these routers have been configured to not respond to echo requests used in the trace. In this example there appears to be a problem between R1 and R2.

A Cisco IOS **traceroute** command output varies from the Windows **tracert** command. For instance, refer to the following topology.

The diagram shows a network topology with the IP addressing of router interfaces and a traceroute command issued from a Cisco IOS router. The topology consists of the following devices and networks, from left to right. A switch on network 192.168.10.0/24 is connected to router R1 at an interface with an address of .1. R1 is connected to router R2 by network 209.165.200.224/30. The interface on R1 has an address of .225 and the interface on R2 has an address of .226. R2 is connected to router R3 by network 209.165.200.228/30. The interface on R2 has an address of .229 and the interface on R3 has an address of .230. R3 is connected to a switch which is connected to host PC B with address 10.1.1.10. R1 is issuing the following trace command from the CLI: R1# traceroute 10.1.1.10.
PC B10.1.1.10192.168.10.0 /24.1R3R1.225.226.229R2.230209.165.200.224 /30209.165.200.228 /30R1

Trace from a Cisco IOS router

```
R1# traceroute 10.1.1.10
```

The following is a sample output of traceroute command from R1.

```
R1# traceroute 10.1.1.10
Type escape sequence to abort.
Tracing the route to 10.1.1.10
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.226 1 msec 0 msec 1 msec
  2 209.165.200.230 1 msec 0 msec 1 msec
  3 10.1.1.10 1 msec 0 msec
R1#
```

In this example, the trace validated that it could successfully reach PC B.

Timeouts indicate a potential problem. For instance, if the 10.1.1.10 host was not available, the **traceroute** command would display the following output.

```
R1# traceroute 10.1.1.10
Type escape sequence to abort.
Tracing the route to 10.1.1.10
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.226 1 msec 0 msec 1 msec
  2 209.165.200.230 1 msec 0 msec 1 msec
  3  *  *  *
  4  *  *  *
  5  *
```

Use **Ctrl-Shift-6** to interrupt a **traceroute** in Cisco IOS.

Note: Windows implementation of traceroute (tracert) sends ICMP Echo Requests. Cisco IOS and Linux use UDP with an invalid port number. The final destination will return an ICMP port unreachable message.

Extended Traceroute

Like the extended **ping** command, there is also an extended **traceroute** command. It allows the administrator to adjust parameters related to the command operation. This is helpful in locating the problem when troubleshooting routing loops, determining the exact next-hop router, or determining where packets are getting dropped or denied by a router or firewall.

The Windows **tracert** command allows the input of several parameters through options in the command line. However, it is not guided like the extended traceroute IOS command. The following output displays the available options for the Windows **tracert** command.

```
C:\Users\PC-A> tracert /?
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w
timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name
Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for
target.
    -j host-list       Loose source route along host-list
(IPv4-only).
    -w timeout         Wait timeout milliseconds for each
reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.
C:\Users\PC-A>
```

The Cisco IOS extended **traceroute** option enables the user to create a special type of trace by adjusting parameters related to the command operation. Extended traceroute is entered in privileged EXEC mode by typing **traceroute** without a destination IP address. IOS will guide you through the command options by presenting a number of prompts related to the setting of all the different parameters.

Note: Pressing **Enter** accepts the indicated default values.

For example, assume you want to test connectivity to PC B from the R1 LAN. Although this could be verified from PC A, an extended **traceroute** could be configured on R1 to specify a different source address.

The diagram shows a network topology with the IP addressing of router interfaces and an extended traceroute command issued from a Cisco IOS router. The topology consists of the following devices and networks, from left to right. A switch on network 192.168.10.0/24 is connected to router R1 at an interface with an address of .1. R1 is connected to router R2 by network 209.165.200.224/30. The interface on R1 has an address of .225 and the interface on R2 has an address of .226. R2 is connected to router R3 by network 209.165.200.228/30. The interface on R2 has an address of .229 and the interface on R3 has an address of .230. R3 is connected to a switch which is connected to host PC B with address 10.1.1.10. R1 is issuing the following trace command from the CLI: R1# traceroute.

PC B192.168.10.0/24209.165.200.224/30209.165.200.228/3010.1.1.10R1R3R2.1.225.226.229.230

Extended trace from a Cisco IOS router

```
R1# traceroute
```

As illustrated in the example, the source IP address of the extended **traceroute** command on R1 could be configured to use the R1 LAN interface IP address (i.e., 192.168.10.1).

```
R1# traceroute
Protocol [ip]:
Target IP address: 10.1.1.10
Ingress traceroute [n]:
Source address: 192.168.10.1
DSCP Value [0]:
Numeric display [n]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Port Number [33434]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 192.168.10.10
VRF info: (vrf in name/id, vrf out name/id)
  1 209.165.200.226 1 msec 1 msec 1 msec
  2 209.165.200.230 0 msec 1 msec 0 msec
  3  *
    10.1.1.10 2 msec 2 msec
R1#
```

17.4.5 Network Baseline

One of the most effective tools for monitoring and troubleshooting network performance is to establish a network baseline. Creating an effective network performance baseline is accomplished over a period of time. Measuring performance at varying times and loads will assist in creating a better picture of overall network performance.

The output derived from network commands contributes data to the network baseline. One method for starting a baseline is to copy and paste the results from an executed **ping**, **trace**, or other relevant commands into a text file. These text files can be time stamped with the date and saved into an archive for later retrieval and comparison.

Among items to consider are error messages and the response times from host to host. If there is a considerable increase in response times, there may be a latency issue to address.

For example, the following **ping** output was captured and pasted into a text file.

August 19, 2019 at 08:14:43

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
```

```
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Reply from 10.1.1.10: bytes=32 time<1ms TTL=64
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\PC-A>
```

Notice the **ping** round-trip times are less than 1 ms.

A month later, the ping is repeated and captured.

September 19, 2019 at 10:18:21

```
C:\Users\PC-A> ping 10.1.1.10
Pinging 10.1.1.10 with 32 bytes of data:
Reply from 10.1.1.10: bytes=32 time=50ms TTL=64
Reply from 10.1.1.10: bytes=32 time=49ms TTL=64
Reply from 10.1.1.10: bytes=32 time=46ms TTL=64
Reply from 10.1.1.10: bytes=32 time=47ms TTL=64
Ping statistics for 10.1.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 46ms, Maximum = 50ms, Average = 48ms
C:\Users\PC-A>
```

Notice this time that the **ping** round-trip times are much longer indicating a potential problem.

Corporate networks should have extensive baselines; more extensive than we can describe in this course. Professional-grade software tools are available for storing and maintaining baseline information. In this course, we cover a few basic techniques and discuss the purpose of baselines.

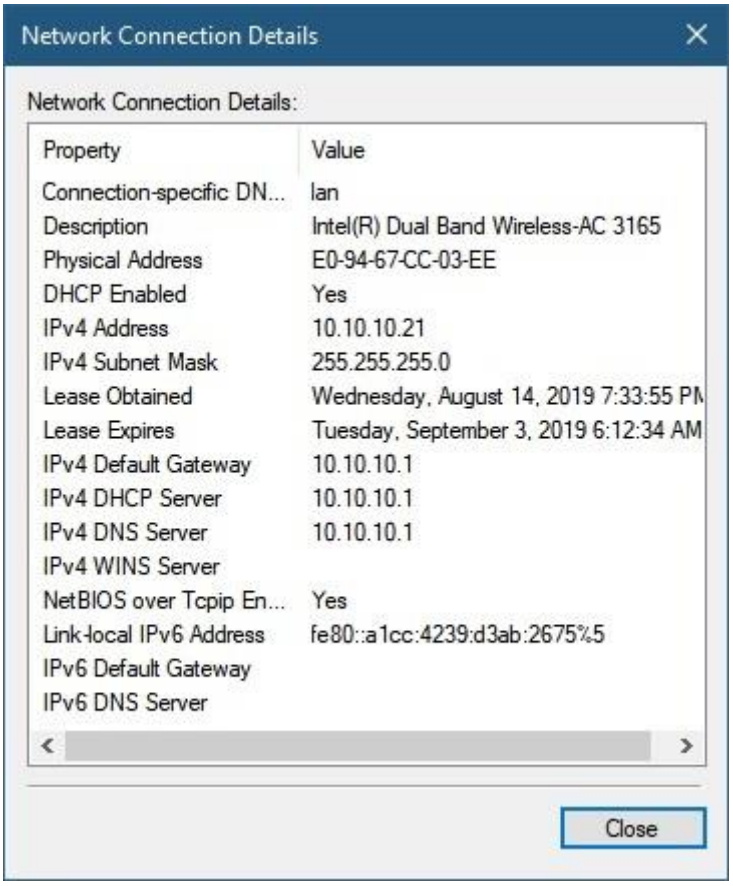
Cisco's best practices for baseline processes can be found by searching the internet for "Baseline Process Best Practices".

5.4.5 Host and IOS Commands

P Configuration on a Windows Host

If you have used any of the tools in the previous topic to verify connectivity and found that some part of your network is not working as it should, now is the time to use some commands to troubleshoot your devices. Host and IOS commands can help you determine if the problem is with the IP addressing of your devices, which is a common network problem.

Checking the IP addressing on host devices is a common practice in networking for verifying and troubleshooting end-to-end connectivity. In Windows 10, you can access the IP address details from the **Network and Sharing Center**, as shown in the figure, to quickly view the four important settings: address, mask, router, and DNS.



However, network administrators typically view the IP addressing information on a Windows host by issuing the **ipconfig** command at the command line of a Windows computer, as shown in the sample output.

```
C:\Users\PC-A> ipconfig
Windows IP Configuration
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16
   IPv4 Address. . . . . . . . . . . : 192.168.10.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1
(Output omitted)
```

Use the **ipconfig /all** command to view the MAC address, as well as a number of details regarding the Layer 3 addressing of the device, as shown in the example output.

```
C:\Users\PC-A> ipconfig /all
Windows IP Configuration
   Host Name . . . . . . . . . . . . : PC-A-00H20
   Primary Dns Suffix  . . . . . . . : cisco.com
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : cisco.com
```

```
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band
Wireless-AC 8265
   Physical Address. . . . . . . . . : F8-94-C2-E4-C5-0A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.10(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : August 17, 2019 1:20:17
PM
   Lease Expires . . . . . . . . . . : August 18, 2019 1:20:18
PM
   Default Gateway . . . . . . . . . : 192.168.10.1
   DHCP Server . . . . . . . . . . . : 192.168.10.1
   DHCPv6 IAID . . . . . . . . . . . : 100177090
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-21-F3-76-75-
54-E1-AD-DE-DA-9A
   DNS Servers . . . . . . . . . . . : 192.168.10.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

If a host is configured as a DHCP client, the IP address configuration can be renewed using the **ipconfig /release** and **ipconfig /renew** commands, as shown in the sample output.

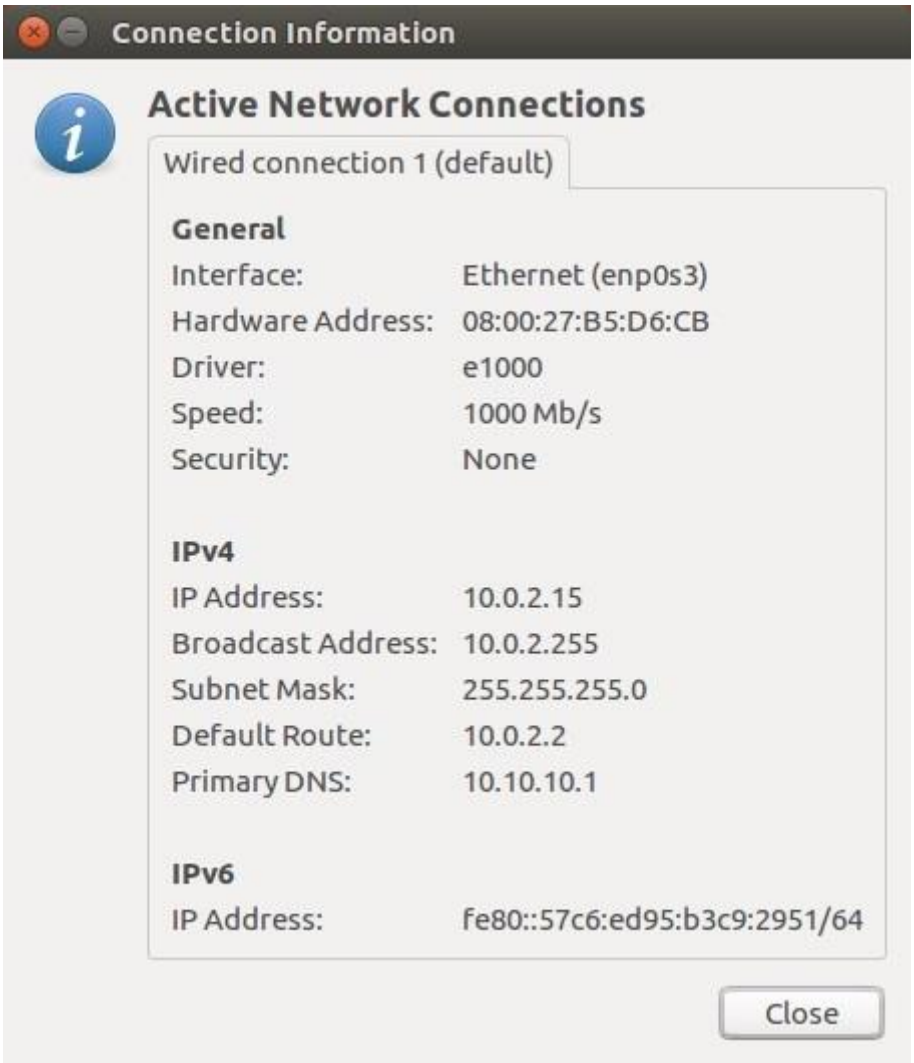
```
C:\Users\PC-A> ipconfig /release
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16
   Default Gateway . . . . . . . . . :
(Output omitted)
C:\Users\PC-A> ipconfig /renew
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16
   IPv4 Address. . . . . . . . . . . : 192.168.1.124
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
(Output omitted)
C:\Users\PC-A>
```

The DNS Client service on Windows PCs also optimizes the performance of DNS name resolution by storing previously resolved names in memory. The **ipconfig /displaydns** command displays all of the cached DNS entries on a Windows computer system, as shown in the example output.

```
C:\Users\PC-A> ipconfig /displaydns
Windows IP Configuration
(Output omitted)
    netacad.com
    ----------------------------------------
    Record Name . . . . . : netacad.com
    Record Type . . . . . : 1
    Time To Live  . . . . : 602
    Data Length . . . . . : 4
    Section . . . . . : Answer
    A (Host) Record . . . : 54.165.95.219
(Output omitted)
```

IP Configuration on a Linux Host

Verifying IP settings using the GUI on a Linux machine will differ depending on the Linux distribution (distro) and desktop interface. The figure shows the **Connection Information** dialog box on the Ubuntu distro running the Gnome desktop.



On the command line, network administrators use the **ifconfig** command to display the status of the currently active interfaces and their IP configuration, as shown in the output.

```
[analyst@secOps ~]$ ifconfig
```

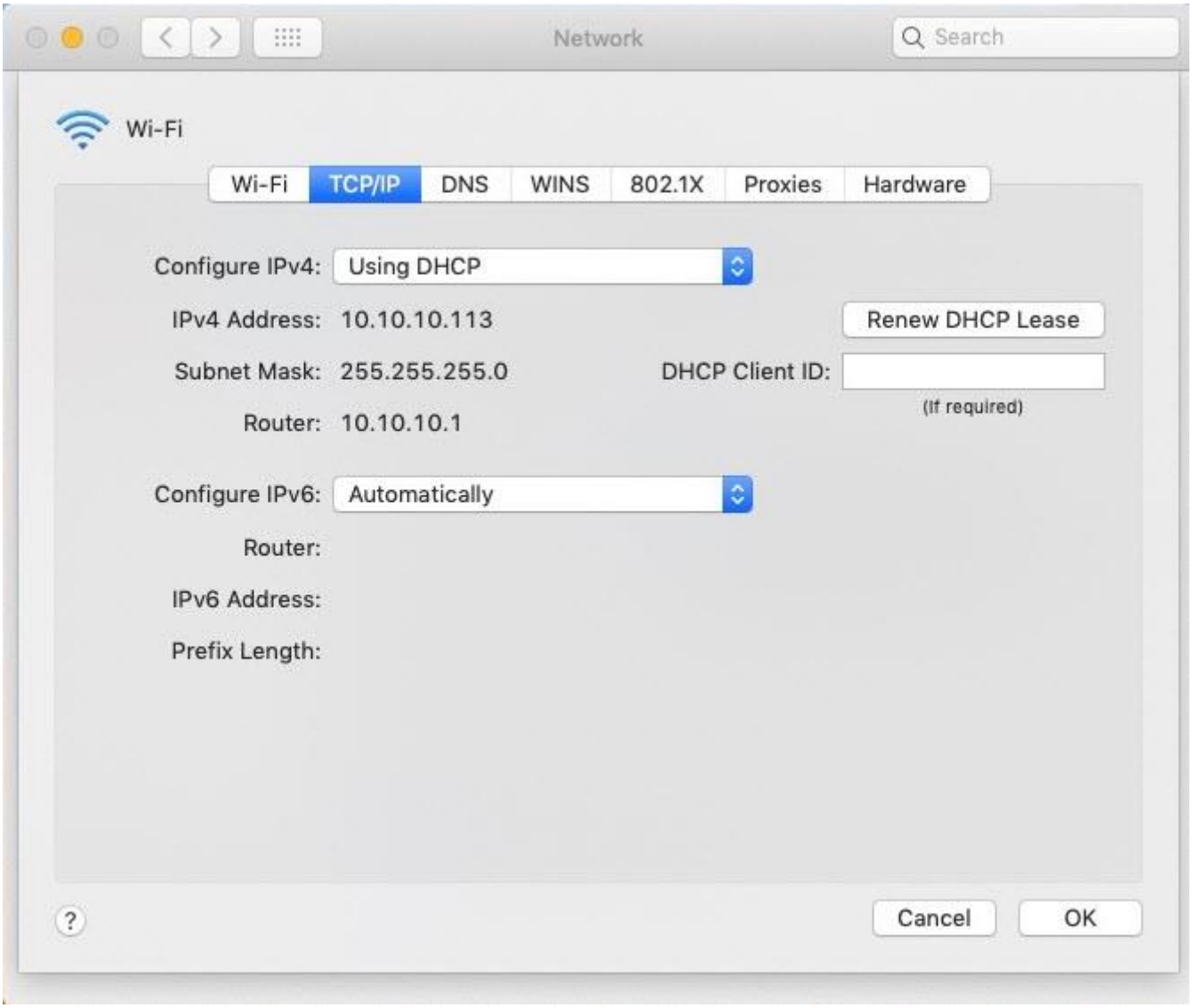
```
enp0s3    Link encap:Ethernet  HWaddr 08:00:27:b5:d6:cb
          inet
addr: 10.0.2.15  Bcast:10.0.2.255  Mask: 255.255.255.0
          inet6 addr: fe80::57c6:ed95:b3c9:2951/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1332239 errors:0 dropped:0 overruns:0
frame:0
          TX packets:105910 errors:0 dropped:0 overruns:0
carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1855455014 (1.8 GB)  TX bytes:13140139 (13.1
MB)
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier
0  collisions 0
```

The Linux **ip address** command is used to display addresses and their properties. It can also be used to add or delete IP addresses.

Note: The output displayed may vary depending on the Linux distribution.

IP Configuration on a macOS Host

In the GUI of a Mac host, open **Network Preferences > Advanced** to get the IP addressing information, as shown in the figure.



However, the **ifconfig** command can also be used to verify the interface IP configuration a shown in the output.

```
MacBook-Air:~ Admin$ ifconfig en0
en0: flags=8863 mtu 1500
        ether c4:b3:01:a0:64:98
        inet6 fe80::c0f:1bf4:60b1:3adb%en0 prefixlen 64 secured
scopeid 0x5
        inet 10.10.10.113 netmask 0xffffff00 broadcast
10.10.10.255
        nd6 options=201
        media: autoselect
        status: active
MacBook-Air:~ Admin$
```

Other useful macOS commands to verify the host IP settings include **networksetup -listallnetworkservices** and the **networksetup -getinfo** <*network service*>**,** as shown in the following output.

```
MacBook-Air:~ Admin$ networksetup -listallnetworkservices
```

```
An asterisk (*) denotes that a network service is disabled.
iPhone USB
Wi-Fi
Bluetooth PAN
Thunderbolt Bridge
MacBook-Air:~ Admin$
MacBook-Air:~ Admin$ networksetup -getinfo Wi-Fi
DHCP Configuration
IP address: 10.10.10.113
Subnet mask: 255.255.255.0
Router: 10.10.10.1
Client ID:
IPv6: Automatic
IPv6 IP address: none
IPv6 Router: none
Wi-Fi ID: c4:b3:01:a0:64:98
MacBook-Air:~ Admin$
```

The arp Command

The **arp** command is executed from the Windows, Linux, or Mac command prompt. The command lists all devices currently in the ARP cache of the host, which includes the IPv4 address, physical address, and the type of addressing (static/dynamic), for each device.

For instance, refer to the topology in the figure.

five hosts with IP addresses 10.0.0.1/24, 10.0.0.2/24, 10.0.0.3/24, 10.0.0.4/24, and 10.0.0.5/24 are connected to a switch connected to a router with an IP address of 10.0.0.254/24 10.0.0.254/2410.0.0.1/2410.0.0.2/2410.0.0.3/2410.0.0.4/2410.0.0.5/24PC-A

The output of the **arp -a** command on the Windows PC-A host is displayed.

```
C:\Users\PC-A> arp -a
Interface: 192.168.93.175 --- 0xc
  Internet Address      Physical Address      Type
  10.0.0.2              d0-67-e5-b6-56-4b     dynamic
  10.0.0.3              78-48-59-e3-b4-01     dynamic
  10.0.0.4              00-21-b6-00-16-97     dynamic
  10.0.0.254            00-15-99-cd-38-d9     dynamic
```

The **arp -a** command displays the known IP address and MAC address binding. Notice how IP address 10.0.0.5 is not included in the list. This is because the ARP cache only displays information from devices that have been recently accessed.

To ensure that the ARP cache is populated, **ping** a device so that it will have an entry in the ARP table. For instance, if PC-A pinged 10.0.0.5, then the ARP cache would contain an entry for that IP address.

The cache can be cleared by using the **netsh interface ip delete arpcache** command in the event the network administrator wants to repopulate the cache with updated information.

Note: You may need administrator access on the host to be able to use the **netsh interface ip delete arpcache** command.

Common show Commands Revisited

In the same way that commands and utilities are used to verify a host configuration, commands can be used to verify the interfaces of intermediary devices. The Cisco IOS provides commands to verify the operation of router and switch interfaces.

The Cisco IOS CLI **show** commands display relevant information about the configuration and operation of the device. Network technicians use **show** commands extensively for viewing configuration files, checking the status of device interfaces and processes, and verifying the device operational status. The status of nearly every process or function of the router can be displayed using a **show** command.

Commonly used **show** commands and when to use them are listed in the table.

Command	Useful for …
show running-config	To verify the current configuration and settings
show interfaces	To verify the interface status and see if there ar
show ip interface	To verify the Layer 3 information of an interfac
show arp	To verify the list of known hosts on the local E
show ip route	To verify the Layer 3 routing information
show protocols	To verify which protocols are operational
show version	To verify the memory, interfaces, and licences

Click the buttons to see example output from each of these show commands.Note: The output of some commands has been edited to focus on pertinent settings and reduce content.
show running-config
show interfaces
show ip interface
show arp
show ip route
show protocols
show version

show running-config

Verifies the current configuration and settings

```
R1# show running-config

(Output omitted)

!

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption
```

```
!
hostname R1
!
interface GigabitEthernet0/0/0
 description Link to R2
 ip address 209.165.200.225 255.255.255.252
 negotiation auto
!
interface GigabitEthernet0/0/1
 description Link to LAN
 ip address 192.168.10.1 255.255.255.0
 negotiation auto
!
router ospf 10
 network 192.168.10.0 0.0.0.255 area 0
 network 209.165.200.224 0.0.0.3 area 0
!
banner motd ^C Authorized access only! ^C
!
line con 0
 password 7 14141B180F0B
 login
line vty 0 4
 password 7 00071A150754
 login
 transport input telnet ssh
!
end
```

```
R1#
```

The show cdp neighbors Command

There are several other IOS commands that are useful. The Cisco Discovery Protocol (CDP) is a Cisco proprietary protocol that runs at the data link layer. Because CDP operates at the data link layer, two or more Cisco network devices, such as routers that support different network layer protocols, can learn about each other even if Layer 3 connectivity has not been established.

When a Cisco device boots, CDP starts by default. CDP automatically discovers neighboring Cisco devices running CDP, regardless of which Layer 3 protocol or suites are running. CDP exchanges hardware and software device information with its directly connected CDP neighbors.

CDP provides the following information about each CDP neighbor device:

- **Device identifiers** - The configured host name of a switch, router, or other device
- **Address list** - Up to one network layer address for each protocol supported
- **Port identifier** - The name of the local and remote port in the form of an ASCII character string, such as FastEthernet 0/0
- **Capabilities list** - For example, whether a specific device is a Layer 2 switch or a Layer 3 switch
- **Platform** - The hardware platform of the device--for example, a Cisco 1841 series router.

Refer to the topology and the **show cdp neighbor** command output.

router R3 is connected via interface G0/0/1 to switch S3 at port F0/5 which is connected to switch S4
G0/0/1F0/5R3S3S4

```
R3# show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source
Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater,
P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay
Device ID        Local
Intrfce     Holdtme    Capability  Platform  Port ID
S3               Gig 0/0/1         122              S I   WS-
C2960+ Fas 0/5
Total cdp entries displayed : 1
R3#
```

The output displays that the R3 GigabitEthernet 0/0/1 interface is connected to the FastEthernet 0/5 interface of S3, which is a Cisco Catalyst 2960+ switch. Notice that R3 has not gathered information about S4. This is because CDP can only discover directly connected Cisco devices. S4 is not directly connected to R3 and therefore is not listed in the output.

The **show cdp neighbors detail** command reveals the IP address of a neighboring device, as shown in the output. CDP will reveal the IP address of the neighbor regardless of whether or not you can ping that neighbor. This command is very helpful when two Cisco routers cannot route across their shared data link. The **show cdp neighbors detail** command will help determine if one of the CDP neighbors has an IP configuration error.

As helpful as CDP is, it can also be a security risk because it can provide useful network infrastructure information to threat actors. For example, by default many IOS versions send CDP advertisements out all enabled ports. However, best practices suggest that CDP should be enabled only on interfaces that are connecting to other infrastructure Cisco devices. CDP advertisements should be disabled on user-facing ports.

Because some IOS versions send out CDP advertisements by default, it is important to know how to disable CDP. To disable CDP globally, use the global configuration command **no cdp run**. To disable CDP on an interface, use the interface command **no cdp enable**.

The show ip interface brief Command

One of the most frequently used commands is the **show ip interface brief** command. This command provides a more abbreviated output than the **show ip interface** command. It provides a summary of the key information for all the network interfaces on a router.

For example, the **show ip interface brief** output displays all interfaces on the router, the IP address assigned to each interface, if any, and the operational status of the interface.

```
R1# show ip interface brief
Interface              IP-Address      OK? Method
Status                Protocol
GigabitEthernet0/0/0   209.165.200.225 YES manual
up                    up
GigabitEthernet0/0/1   192.168.10.1    YES manual
up                    up
Serial0/1/0            unassigned      NO  unset  down
         down
Serial0/1/1            unassigned      NO  unset  down
         down
GigabitEthernet0       unassigned      YES
unset  administratively down down
R1#
```

Verify Switch Interfaces

The **show ip interface brief** command can also be used to verify the status of the switch interfaces, as shown in the output.

```
S1# show ip interface brief
Interface              IP-Address      OK? Method
Status                Protocol
Vlan1                  192.168.254.250 YES manual
up                    up
FastEthernet0/1        unassigned      YES
unset  down                  down
```

```
FastEthernet0/2        unassigned      YES
unset  up                    up
FastEthernet0/3        unassigned      YES
unset  up                    up
```

The VLAN1 interface is assigned an IPv4 address of 192.168.254.250, has been enabled, and is operational.

The output also shows that the FastEthernet0/1 interface is down. This indicates that either no device is connected to the interface or the device that is connected has a network interface that is not operational.

In contrast, the output shows that the FastEthernet0/2 and FastEthernet0/3 interfaces are operational. This is indicated by both the status and protocol being shown as up.

5.4.6 Troubleshooting Methodologies

Basic Troubleshooting Approaches

In the previous two topics, you learned about some utilities and commands that you can use to help identify problem areas in your network. This is an important part of troubleshooting. There are many ways to troubleshoot a network problem. This topic details a structured troubleshooting process that can help you to become a better network administrator. It also provides a few more commands to help you resolve problems. Network problems can be simple or complex, and can result from a combination of hardware, software, and connectivity issues. Technicians must be able to analyze the problem and determine the cause of the error before they can resolve the network issue. This process is called troubleshooting.

A common and efficient troubleshooting methodology is based on the scientific method.

The table shows the six main steps in the troubleshooting process.

Step	Description
Step 1. Identify the Problem	• This is the first step in the troubleshooting process. • Although tools can be used in this step, a conversation with the user is often very helpful.
Step 2. Establish a Theory of Probable Causes	• After the problem is identified, try to establish a theory of probable causes. • This step often yields more than a few probable causes to the problem.
Step 3. Test the Theory to Determine Cause	• Based on the probable causes, test your theories to determine which one is the cause of the problem. • A technician will often apply a quick procedure to test and see if it solves the problem.

- If a quick procedure does not correct the problem, you might need to research the problem further to establish the exact cause.

Step 4. Establish a Plan of Action and Implement the Solution	After you have determined the exact cause of the problem, establish a plan of action to resolve the problem and implement the solution.
Step 5. Verify Solution and Implement Preventive Measures	• After you have corrected the problem, verify full functionality. • If applicable, implement preventive measures.
Step 6. Document Findings, Actions, and Outcomes	• In the final step of the troubleshooting process, document your findings, actions, and outcomes. • This is very important for future reference.

To assess the problem, determine how many devices on the network are experiencing the problem. If there is a problem with one device on the network, start the troubleshooting process at that device. If there is a problem with all devices on the network, start the troubleshooting process at the device where all other devices are connected. You should develop a logical and consistent method for diagnosing network problems by eliminating one problem at a time.

Resolve or Escalate?

In some situations, it may not be possible to resolve the problem immediately. A problem should be escalated when it requires a manager decision, some specific expertise, or network access level unavailable to the troubleshooting technician.

For example, after troubleshooting, the technician concludes a router module should be replaced. This problem should be escalated for manager approval. The manager may have to escalate the problem again as it may require the approval of the financial department before a new module can be purchased.

A company policy should clearly state when and how a technician should escalate a problem.

The debug Command

OS processes, protocols, mechanisms and events generate messages to communicate their status. These messages can provide valuable information when troubleshooting or verifying system operations. The IOS **debug** command allows the administrator to display these messages in real-time for analysis. It is a very important tool for monitoring events on a Cisco IOS device.

All **debug** commands are entered in privileged EXEC mode. The Cisco IOS allows for narrowing the output of **debug** to include only the relevant feature or subfeature. This is important because debugging output is assigned high priority in the CPU process and it can render the system unusable. For this reason, use **debug** commands only to troubleshoot specific problems.

For example, to monitor the status of ICMP messages in a Cisco router, use **debug ip icmp**, as shown in the example.

```
R1# debug ip icmp
ICMP packet debugging is on
R1#
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
R1#
*Aug 20 14:18:59.605: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.606: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.608: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.609: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
*Aug 20 14:18:59.611: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
R1#
```

To list a brief description of all the debugging command options, use the **debug ?** command in privileged EXEC mode at the command line.

To turn off a specific debugging feature, add the **no** keyword in front of the **debug** command:

```
Router# no debug ip icmp
```

Alternatively, you can enter the **undebug** form of the command in privileged EXEC mode:

```
Router# undebug ip icmp
```

To turn off all active debug commands at once, use the **undebug all** command:

```
Router# undebug all
```

Be cautious using some **debug** command. Commands such as **debug all** and **debug ip packet** generate a substantial amount of output and can use a large portion of system resources. The router could get so busy displaying **debug** messages that it would not have enough processing power to perform its network functions, or even listen to commands to turn off debugging. For this reason, using these command options is not recommended and should be avoided.

The terminal monitor Command

Connections to grant access to the IOS command line interface can be established in the following two ways:

- **Locally** - Local connections (i.e., console connection) require physical access to the router or switch console port using a rollover cable.
- **Remotely** - Remote connections require the use of Telnet or SSH to establish a connection to an IP configured device.

Certain IOS messages are automatically displayed on a console connection but not on a remote connection. For instance, **debug** output is displayed by default on console connections. However, **debug** output is not automatically displayed on remote connections. This is because **debug** messages are log messages which are prevented from being displayed on vty lines.

In the following output for instance, the user established a remote connection using Telnet from R2 to R1. The user then issued the **debug ip icmp** command. However, the command failed to display **debug** output.

```
R2# telnet 209.165.200.225
Trying 209.165.200.225 ... Open
Authorized access only!
User Access Verification
Password:
R1> enable
Password:
R1# debug ip icmp
ICMP packet debugging is on
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
R1#
! No debug output displayed>
```

To display log messages on a terminal (virtual console), use the **terminal monitor** privileged EXEC command. To stop logging messages on a terminal, use the **terminal no monitor** privileged EXEC command.

For instance, notice how the **terminal monitor** command has now been entered and the **ping** command displays the **debug** output.

```
R1# terminal monitor
R1# ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/2 ms
R1#
*Aug 20 16:03:49.735: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
```

```
**Aug 20 16:03:49.737: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.738: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.740: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
**Aug 20 16:03:49.741: ICMP: echo reply rcvd, src 10.1.1.1, dst
209.165.200.225,topology BASE, dscp 0 topoid 0
R1# no debug ip icmp
ICMP packet debugging is off
R1#
```

Note: The intent of the **debug** command is to capture live output for a short period of time (i.e., a few seconds to a minute or so). Always disable **debug** when not required.

5.4.7 Troubleshooting Scenarios

Duplex Operation and Mismatch Issues

Many common network problems can be identified and resolved with little effort. Now that you have the tools and the process for troubleshooting a network, this topic reviews some common networking issues that you are likely to find as a network administrator.

In data communications, *duplex* refers to the direction of data transmission between two devices.

There are two duplex communication modes:

- **Half-duplex -** Communication is restricted to the exchange of data in one direction at a time.
- **Full-duplex -** Communications is permitted to be sent and received simultaneously.

The figure illustrates how each duplex method operates.

The figure is a comparison of half-duplex versus full-duplex communications. The top diagram shows half-duplex communication. Switch S1 is connected to switch S2 with an arrow flowing from S1 to S2 indicating a device can send or receive. The bottom diagram shows-full duplex communication. Switch S1 is connected to switch S2 with arrows pointing in both directions indicating a device can send AND receive simultaneously.
S2S1S2S1
Full-Duplex Communication
Half-Duplex Communication

Send **OR** receive
Send **AND** receive simultaneously

Interconnecting Ethernet interfaces must operate in the same duplex mode for best communication performance and to avoid inefficiency and latency on the link.

The Ethernet autonegotiation feature facilitates configuration, minimizes problems and maximizes link performance between two interconnecting Ethernet links. The connected devices first announce their supported capabilities and then choose the highest performance

mode supported by both ends. For example, the switch and router in the figure have successfully autonegotiated full-duplex mode.

S1R1

F0/5

G0/0/1

If one of the two connected devices is operating in full-duplex and the other is operating in half-duplex, a duplex mismatch occurs. While data communication will occur through a link with a duplex mismatch, link performance will be very poor.

Duplex mismatches are typically caused by a misconfigured interface or in rare instances by a failed autonegotiation. Duplex mismatches may be difficult to troubleshoot as the communication between devices still occurs.

IP Addressing Issues on IOS Devices

IP address-related problems will likely keep remote network devices from communicating. Because IP addresses are hierarchical, any IP address assigned to a network device must conform to that range of addresses in that network. Wrongly assigned IP addresses create a variety of issues, including IP address conflicts and routing problems.

Two common causes of incorrect IPv4 assignment are manual assignment mistakes or DHCP-related issues.

Network administrators often have to manually assign IP addresses to devices such as servers and routers. If a mistake is made during the assignment, then communications issues with the device are very likely to occur.

On an IOS device, use the **show ip interface** or **show ip interface brief** commands to verify what IPv4 addresses are assigned to the network interfaces. For example, issuing the **show ip interface brief** command as shown would validate the interface status on R1.

```
R1# show ip interface brief
Interface              IP-Address      OK? Method
Status                Protocol
GigabitEthernet0/0/0   209.165.200.225 YES manual
up                    up
GigabitEthernet0/0/1   192.168.10.1    YES manual
up                    up
Serial0/1/0            unassigned      NO  unset  down
         down
Serial0/1/1            unassigned      NO  unset  down
         down
GigabitEthernet0       unassigned      YES
unset  administratively down down
R1#
```

IP Addressing Issues on End Devices

In Windows-based machines, when the device cannot contact a DHCP server, Windows will automatically assign an address belonging to the 169.254.0.0/16 range. This feature is called Automatic Private IP Addressing (APIPA) and is designed to facilitate communication within the local network. Think of it as Windows saying, “I will use this address from the 169.254.0.0/16 range because I could not get any other address”.

Often, a computer with an APIPA address will not be able to communicate with other devices in the network because those devices will most likely not belong to the 169.254.0.0/16 network. This situation indicates an automatic IPv4 address assignment problem that should be fixed.

Note: Other operating systems, such Linux and OS X, will not assign an IPv4 address to the network interface if communication with a DHCP server fails.

Most end devices are configured to rely on a DHCP server for automatic IPv4 address assignment. If the device is unable to communicate with the DHCP server, then the server cannot assign an IPv4 address for the specific network and the device will not be able to communicate.

To verify the IP addresses assigned to a Windows-based computer, use the **ipconfig** command, as shown in the output.

```
C:\Users\PC-A> ipconfig
Windows IP Configuration
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16
   IPv4 Address. . . . . . . . . . . : 192.168.10.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1
(Output omitted)
```

Default Gateway Issues

The default gateway for an end device is the closest networking device that can forward traffic to other networks. If a device has an incorrect or nonexistent default gateway address, it will not be able to communicate with devices in remote networks. Because the default gateway is the path to remote networks, its address must belong to the same network as the end device.

The address of the default gateway can be manually set or obtained from a DHCP server. Similar to IPv4 addressing issues, default gateway problems can be related to misconfiguration (in the case of manual assignment) or DHCP problems (if automatic assignment is in use).

To solve misconfigured default gateway issues, ensure that the device has the correct default gateway configured. If the default address was manually set but is incorrect, simply replace it with the proper address. If the default gateway address was automatically set, ensure the device can communicate with the DHCP server. It is also important to verify that the proper

IPv4 address and subnet mask were configured on the interface of the router and that the interface is active.

To verify the default gateway on Windows-based computers, use the **ipconfig** command as shown.

```
C:\Users\PC-A> ipconfig
Windows IP Configuration
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16
   IPv4 Address. . . . . . . . . . . : 192.168.10.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.10.1
(Output omitted)
```

On a router, use the **show ip route** command to list the routing table and verify that the default gateway, known as a default route, has been set. This route is used when the destination address of the packet does not match any other routes in its routing table.

For example, the output verifies that R1 has a default gateway (i.e., Gateway of last resort) configured pointing to IP address 209.168.200.226.

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.226 to network 0.0.0.0
O*E2  0.0.0.0/0 [110/1] via 209.165.200.226, 02:19:50,
GigabitEthernet0/0/0
      10.0.0.0/24 is subnetted, 1 subnets
O        10.1.1.0 [110/3] via 209.165.200.226, 02:05:42,
GigabitEthernet0/0/0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.10.1/32 is directly connected,
GigabitEthernet0/0/1
      209.165.200.0/24 is variably subnetted, 3 subnets, 2
masks
C        209.165.200.224/30 is directly connected,
GigabitEthernet0/0/0
L        209.165.200.225/32 is directly connected,
GigabitEthernet0/0/0
O        209.165.200.228/30
           [110/2] via 209.165.200.226, 02:07:19,
GigabitEthernet0/0/0
R1#
```

The first highlighted line basically states that the gateway to any (i.e., 0.0.0.0) should be sent to IP address 209.165.200.226. The second highlighted displays how R1 learned about the default gateway. In this case, R1 received the information from another OSPF-enabled router.

Troubleshooting DNS Issues

Domain Name System (DNS) defines an automated service that matches names, such as www.cisco.com, with the IP address. Although DNS resolution is not crucial to device communication, it is very important to the end user.

It is common for users to mistakenly relate the operation of an internet link to the availability of the DNS. User complaints such as “the network is down” or “the internet is down” are often caused by an unreachable DNS server. While packet routing and all other network services are still operational, DNS failures often lead the user to the wrong conclusion. If a user types in a domain name such as www.cisco.com in a web browser and the DNS server is unreachable, the name will not be translated into an IP address and the website will not display.

DNS server addresses can be manually or automatically assigned. Network administrators are often responsible for manually assigning DNS server addresses on servers and other devices, while DHCP is used to automatically assign DNS server addresses to clients.

Although it is common for companies and organizations to manage their own DNS servers, any reachable DNS server can be used to resolve names. Small office and home office (SOHO) users often rely on the DNS server maintained by their ISP for name resolution. ISP-maintained DNS servers are assigned to SOHO customers via DHCP. Additionally, Google maintains a public DNS server that can be used by anyone and it is very useful for testing. The IPv4 address of Google’s public DNS server is 8.8.8.8 and 2001:4860:4860::8888 for its IPv6 DNS address.

Cisco offers OpenDNS which provides secure DNS service by filtering phishing and some malware sites. You can change your DNS address to 208.67.222.222 and 208.67.220.220 in the Preferred DNS server and Alternate DNS server fields. Advanced features such as web content filtering and security are available to families and businesses.

Use the **ipconfig /all** as shown to verify which DNS server is in use by the Windows computer.

```
C:\Users\PC-A> ipconfig /all
(Output omitted)
Wireless LAN adapter Wi-Fi:
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band
Wireless-AC 8265
   Physical Address. . . . . . . . . : F8-94-C2-E4-C5-0A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . :
fe80::a4aa:2dd1:ae2d:a75e%16(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.10.10(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : August 17, 2019 1:20:17
PM
   Lease Expires . . . . . . . . . . : August 18, 2019 1:20:18
PM
   Default Gateway . . . . . . . . . : 192.168.10.1
```

```
   DHCP Server . . . . . . . . . . . . : 192.168.10.1
   DHCPv6 IAID . . . . . . . . . . . . : 100177090
   DHCPv6 Client DUID. . . . . . . . . : 00-01-00-01-21-F3-76-75-
54-E1-AD-DE-DA-9A
   DNS Servers . . . . . . . . . . . . : 208.67.222.222
   NetBIOS over Tcpip. . . . . . . . . : Enabled
(Output omitted)
```

The **nslookup** command is another useful DNS troubleshooting tool for PCs.
With **nslookup** a user can manually place DNS queries and analyze the DNS response.
The **nslookup** command shows the output for a query for www.cisco.com. Notice you can also simply enter an IP address and **nslookup** will resolve the name.

Note: It is not always possible to type an IP address in **nslookup** and receive the domain name. One of the most common reasons for this is that most websites run on servers that support multiple sites.

```
C:\Users\PC-A> nslookup
Default Server:  Home-Net
Address:  192.168.1.1
> cisco.com
Server:  Home-Net
Address:  192.168.1.1
Non-authoritative answer:
Name:    cisco.com
Addresses:  2001:420:1101:1::185
          72.163.4.185
> 8.8.8.8
Server:  Home-Net
Address:  192.168.1.1
Name:    dns.google
Address:  8.8.8.8
>
> 208.67.222.222
Server:  Home-Net
Address:  192.168.1.1
Name:    resolver1.opendns.com
Address:  208.67.222.222
>
```